

TrayGuard

A Local Tray Training And Assessment System

Final Design Review

EC ENGR 180DW – Systems Design

University of California, Los Angeles

Prepared by: Owen Fan, Matthew Li, Andy Ma, Jacob Moore, Qiyu Xin

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Note: Each section of this report lists in parentheses the team member(s) who authored that section, following the FDR report outline convention.

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1 Executive Summary

TODO: Write this section last after the rest of the report is complete. It should summarize the final problem, the selected TrayGuard direction, the strongest evidence, the main design decisions, what was built, the current limitations, and the recommended future validation path.

2 Introduction And Project Overview

Tray content errors during surgical instrument reconstruction are recurring operational failures that can delay surgery and compromise patient safety. The workforce faces structural pressure from growing demand, high turnover, and limited training capacity. Current training blends online coursework, classroom instruction, and hands-on externships that are capacity-constrained by clinical-site availability, preceptor attention, and local teaching materials.

TrayGuard originally targeted a computer-vision-based tray checker: a camera above the assembly workstation would detect each instrument as the technician placed it, compare detections against the tray list, and return a go or no-go verdict before sealing. The CV approach failed early because it required the project to validate detection accuracy, lighting tolerance, occluded-item handling, and workflow integration inside a live SPD environment, resources not available within our project scope. The project pivoted to a training-centered alternative that preserves the same error categories (missing, wrong, extra, misidentified, and miscounted items) but measures them in a simulated local tray task using AprilTag instrument cards, retrieval study, and feedback modes.

This Functional Design Review serves as the literature-grounded design handoff for a future implementation team. It defines the problem, justifies the training-centered solution, specifies requirements and system architecture, provides a study plan for novice evaluation, and documents the scope boundaries that prevent overclaiming.

3 Problem Definition And Scope

3.1 Precise Problem Statement

Sterile processing technicians receive used instruments from the OR, clean, inspect, assemble procedure-specific trays, package, sterilize, store, and distribute sets for surgical use. Mistakes can occur at several points in that workflow: a technician may confuse two visually similar instruments, place the wrong size or pattern into a tray, omit or add an instrument, use the wrong count sheet, or fail to catch a documentation problem. In the OR, those mistakes appear as missing, wrong, extra, damaged, contaminated, or otherwise unavailable instruments.

Published rates vary by hospital, measurement method, and error definition, but the available evidence suggests that these are recurring operational failures, not isolated accidents. Alfred et al. analyzed 3,900 tray defects across 41,799 cases, or roughly 9 defects per 100 cases, and found that 55.0% of recorded defects occurred during assembly [1]. Zhu et al. found 398 packaging errors among 33,839 surgical instrument packages, or about 1.2% of packages, including wrong specifications, incomplete packages, and missing instruments [2]. Nichol et al. observed 236 surgical instrument errors affecting 147 cases, with most errors tied to visualization work such as inspection, identification, function checking, and tray assembly [3].

The general problem is therefore:

Sterile processing workflows do not always reliably produce complete, correct, functional, sterile, and ready-for-use surgical instrument sets.

3.2 Practice Context

The problem appears across the reusable stainless surgical instrument chain. A simplified SPD-to-OR cycle includes the following stages [4]:

1. Point-of-use preparation: at the end of a procedure, instruments are prepared for transfer from the OR to sterile processing. This can include gross cleaning, use of sterile water, and enzymatic pretreatment to keep material from drying on instruments.
2. Transport to sterile processing: used instruments are moved from the OR or procedural area to the sterile processing unit.
3. Decontamination: instruments are cleaned and decontaminated. George et al. describe manual scrubbing, ultrasonic cleaning, and automated washer systems as common parts of this stage.
4. Assembly and inspection: after decontamination, instruments move to the assembly area. Technicians inspect instruments, assemble instrument pans or sets, replace missing items, and place instruments into containers or wraps for sterilization.
5. Sterilization: packaged instruments are sterilized using methods appropriate to the device and material. George et al. describe high-temperature methods such as autoclave sterilization and low-temperature methods such as chemical or radiation-based sterilization.
6. Storage: once sterilized, instruments and trays are cooled as needed and placed in storage until they are required again.
7. Return to use: when needed for surgery, stored instruments are obtained and brought back to the OR for the next case.

Nichol and Saari’s risk model provides a more detailed version of this cycle: for simple instruments, they mapped 104 human tasks and found that 91% of modeled error risk occurred during the sterile-processing portion of the cycle, especially the decontamination and assembly-and-inspection steps [5]. Those steps are high risk because they contain the most complicated, non-routine tasks: instruments must be separated, cleaned, inspected for bioburden and damage, identified, sorted, and rebuilt into the correct tray, and all eight highest-risk tasks in the model involved human visualization, inspection, identification, or sorting.

3.3 Narrowed Problem

Within the full SPD-to-OR cycle, this project narrows to tray reconstruction and verification: the point where cleaned instruments are inspected, identified, matched to count sheets, and rebuilt into procedure-specific trays. This is the most defensible focus because the highest-risk parts of the cycle depend on human visualization, sorting, and local tray knowledge. This focus also addresses a likely stakeholder objection. Even if a tray comes from a specific surgery and is expected to return with the same contents, it does not remain a sealed, unchanged kit during reprocessing. Instruments are transported to decontamination, opened, separated, disassembled when required, manually cleaned or transferred onto washer racks, and exposed to mechanical washing before moving to the clean preparation area [6], [7], [8], [9]. After cleaning, technicians manually rebuild sets against tray-specific count sheets, creating opportunities for items to be mixed between sets, left behind, substituted from backup inventory, or misidentified because of incomplete count-sheet data, local jargon, or look-alike instruments [10], [1], [11].

3.4 Stakeholders And Importance

Instrument readiness matters because surgery is time-sensitive, coordination-heavy work. When an instrument set is wrong, incomplete, unclean, damaged, or unavailable, the OR team may need to pause,

search for replacement instruments, open backup trays, change the procedure flow, or delay the case [3], [12]. Dirty or damaged instruments add a direct safety concern. A 2024 hospital inspection reported pitting, stains, sticky residue, rust, scratches, and other concerns in randomly selected ready-for-use trays [13]. The Pennsylvania Patient Safety Authority warns that contaminated instruments can put patients at risk of surgical site infection and can also cause lost OR time if discovered after a procedure has begun [12]. The stakeholders include:

- Patients, who depend on sterile, functional instruments during surgery.
- Surgeons and OR teams, who need complete and usable sets to maintain procedure flow.
- SPD technicians, who perform detailed, high-volume work under production pressure.
- SPD supervisors and educators, who manage training, quality assurance, staffing, documentation, and feedback loops.
- Hospitals, which lose time and capacity when instrument issues delay cases or require rework.

3.5 General Interest Beyond One Local Practice

This problem is broadly relevant because reusable surgical instruments, SPDs, procedure-specific trays, count sheets, sterilization records, and OR dependence on prepared sets are common across hospitals. The exact instruments, tray lists, vendor systems, and surgeon preferences vary by site, while the readiness challenge repeats across settings: hospitals must transform complex, locally defined instrument requirements into complete and ready surgical sets. The local tray-reconstruction problem connects to broader work on surgical tray optimization, inventory visibility, and instrument-set readiness. dos Santos et al. describe tray rationalization as deciding which instruments and quantities belong in which trays and which trays are needed for procedures [14]. A 2025 industry benchmark report summarized by Healthcare Purchasing News found that 58% of surveyed SPD and OR leaders reported surgery delays caused by instrument sets that were missing, incomplete, or not ready on time [15]. The same benchmark report emphasized human error, inventory gaps, limited data visibility, and real-time visibility into tray and instrument location as major readiness concerns [16]. This makes the problem generalizable without making it generic: each hospital has its own instruments and tray rules, but many hospitals face the same underlying need to verify that a locally defined set is complete, correct, clean, functional, and available before surgery.

4 Preliminary Analysis

The following analysis tools were applied to the training-centered solution path. They document how the problem was decomposed into design objectives, how the system function was modeled, what performance targets were set, how stakeholder needs mapped to engineering characteristics, what solution alternatives were generated, and how the training path was selected.

4.1 Objective Tree Analysis

The objective tree decomposes the top-level goal, “Improve novice readiness for local tray reconstruction through repeatable, measurable simulated practice,” into the functional, measurable, instructor-support, and claim-boundary objectives that guided the design process.

1. Increase local tray knowledge
 - 1.1 Teach local names, aliases, and instrument families
 - 1.2 Teach required counts and tray-specific rules
 - 1.3 Teach lookalike and same-family distinctions
2. Improve applied tray-sorting performance
 - 2.1 Reduce missing required items

- 2.2 Reduce wrong and misidentified items
- 2.3 Reduce extra and wrong-count items
- 2.4 Maintain reasonable task time without rewarding guessing
- 3. Make learning measurable
 - 3.1 Capture pre-test and post-test accuracy
 - 3.2 Export item-level error categories
 - 3.3 Capture confidence, Not sure choices, and high-confidence errors
 - 3.4 Preserve comparable assessment variants
- 4. Support instructors
 - 4.1 Show weak instruments and weak tray rules
 - 4.2 Keep modules local, versioned, and reviewable
 - 4.3 Avoid collecting patient data or unnecessary learner identifiers
- 5. Preserve claim boundaries
 - 5.1 Label outputs as simulated practice evidence
 - 5.2 Require expert review before content-valid claims
 - 5.3 Require SPD pilot evidence before workplace-effectiveness claims

The tree separates objectives into tiers that drove subsequent analysis: Tier 1 (knowledge) informed the retrieval-first study card design; Tier 2 (performance) informed the simulated tray-sort and error-category scoring; Tier 3 (measurability) informed the pre/post assessment architecture; Tier 4 (instructor support) informed the export and reporting subsystem; Tier 5 (claim boundaries) informed the report-labeling and scope-statement requirements.

4.2 Function Analysis – Transparent Box Model

The training system is modeled as a transparent box that transforms a local tray module and physical instrument cards into learner performance evidence through sequential subfunctions:

Table 1: Transparent box model for the training system

Input	Subfunction	Output
Local tray module (JSON)	Authoring and verification	Verified tray template with required items, distractors, lookalike pairs, and assessment variants
Verified tray template	Pre-test assessment	Baseline accuracy, error-category counts, and confidence distribution
Instrument cards and tray template	Retrieval-first study	Prompt-before-reveal card practice with distinguishing features and aliases
Study session output	Identification quiz	Quiz accuracy, duration, and item-level confidence per instrument concept
Cards and tray template	Simulated tray sorting	Selected instrument set scored against required counts
Scored tray selection	Feedback generation	Missing, extra, wrong, misidentified, and wrong-count error report
Pre-test and post-test scores	Metrics aggregation	Pre/post accuracy gain, error reduction, confidence calibration, and weak-item report
Aggregated metrics	Export and reporting	CSV/JSON export with attempt-level and item-level records

The transparent box model separates the learning workflow into discrete subfunctions, each of which was independently analyzed for solution alternatives and performance targets.

4.3 Performance Specification Analysis

Performance specifications define measurable thresholds for each subfunction output. The targets are deliberately formative rather than clinical: they define when the prototype is ready to support a bounded novice-learning claim.

Table 2: Performance specifications for the training system

Specification	Target	Derived from
Module completion time	Pre/post loop finishable in 45–60 minutes	Objective 2.4 (reasonable task time)
Pre/post accuracy gain	Mean gain ≥ 20 percentage points	Objective 3.1 (measurable improvement)
Severe-error reduction	Missing+wrong+misidentified decrease $\geq 30\%$	Objective 2.1, 2.2 (error reduction)
Post-test floor	Mean accuracy $\geq 80\%$	Objective 3.1 (usable simulated familiarity)
Marker detection rate	$\geq 95\%$ in normal layouts	Objective 4.2 (reliable measurement tool)
Export completeness	100% of completed runs export all required fields	Objective 4.1 (instructor-visible evidence)

Full performance specifications with evidence basis and measurement methods are documented in the requirements definition.

4.4 Quality Function Deployment

The QFD matrix maps stakeholder needs to engineering characteristics, target values, and verification methods for the training system.

Table 3: Quality function deployment: training path

Stakeholder need	Engineering characteristic	Target value	Verification
Learners repeat practice with local tray content	File-backed module with items, distractors, aliases, variants	One complete <code>basic_general_tray_v1</code> module	Module schema review and demo load
Practice resembles real tray reconstruction	Physical card sorting with tray-rule scoring	Pre/post/delayed all use same tray rule	End-to-end run
Errors map to meaningful categories	Scoring: missing, extra, wrong, misidentified, wrong-count	Every tray scored with category counts	Unit tests
Assessment is not contaminated by coaching	Mode boundaries with feedback suppression	No hints in pre-test, post-test, or delayed modes	UI walkthrough
Instructors see learner weaknesses	Attempt-level and item-level export	100% export completion	Schema validation
Marker errors do not corrupt scores	High marker reliability and user confirmation	$\geq 95\%$ detection rate	Marker log
System is ethical and nonpunitive	Privacy: hashed IDs, no PHI, report labels	No patient data, labels say practice evidence	Report review

A detailed QFD with evidence-source citations appears in the requirements definition.

4.5 Morphological Chart

The morphological chart generates solution alternatives for each subfunction identified in the transparent box model.

Table 4: Morphological chart: solution alternatives by subfunction

Subfunction	Alternative 1	Alternative 2	Alternative 3	Alternative 4
Content delivery	File-backed JSON module	Online database	Spreadsheet import	Hard-coded tray lists
Instrument representation	Physical marker cards	On-screen instrument icons	Real instruments	Text-based item list
Marker technology	AprilTag	QR code	ArUco	Manual selection
Learner interaction	Study + quiz + tray sort	Study-only	Quiz-only	Tray-sort only
Assessment mode	Pre/post with matched variants	Single post-test	Pre/post without variants	Continuous assessment
Feedback mechanism	Error-category report	Pass/fail only	Item-level hints	Instructor review only
Scoring approach	Category-based scoring (missing, extra, wrong, misidentified, wrong-count)	Simple accuracy (correct/total)	Weighted error severity	Confidence-adjusted scoring
Export format	CSV + JSON per run	Single PDF report	Instructor dashboard	Raw database access

The selected combination is: file-backed JSON module, physical AprilTag marker cards, full study-quiz-sort learning loop, pre/post matched-variant assessment, category-based scoring, error-category feedback, and CSV/JSON export. The alternatives document the deliberate choices and rejected paths that led to this configuration.

4.6 Weighted Objective Method

The weighted objective method scores each candidate solution combination against weighted criteria derived from the objective tree and stakeholder needs.

Table 5: Weighted objective method: solution path selection

Criterion	Weight	Training path	CV tray checker	Software-only quiz	Real-instrument CV
Buildability within project scope	0.25	5	1	4	1
Measurable learning evidence	0.20	5	2	3	3
Physical task fidelity	0.15	3	5	1	5
Low validation burden	0.15	4	1	4	1
Instructor usefulness	0.10	4	3	3	3
Future extensibility	0.10	3	4	2	4
Cost and equipment requirements	0.05	5	3	5	2
Weighted total	1.00	4.35	2.40	3.15	2.60

The training path scores highest (4.35/5.00), driven by buildability, measurable evidence, and low validation burden. The CV tray checker scores lowest on buildability and validation burden because deployment-grade instrument recognition requires local data collection, cross-manufacturer generaliza-

tion testing, clinical workflow integration, and failure-mode analysis that exceed the project timeline. The software-only quiz scores well on buildability but loses points on physical task fidelity. The real-instrument CV path combines the weaknesses of the CV tray checker with additional equipment and handling requirements.

Table 6: Weighted objective method: training path scoring detail

Criterion	Rationale for score
Buildability (5/5)	File-backed module, printable cards, and local app are implementable with available resources within one quarter.
Measurable evidence (5/5)	Pre/post accuracy, error categories, confidence, and export provide structured evidence without requiring clinical deployment.
Physical task fidelity (3/5)	Marker cards preserve spatial search and physical selection but are less realistic than real instruments. Acceptable for a simulated novice study.
Low validation burden (4/5)	Validation requires novice study protocol, expert review, and marker bench test – all within project scope.
Instructor usefulness (4/5)	Export and weak-item reporting support instructor review, but SPD-specific integration is deferred.
Future extensibility (3/5)	Modular design supports future CV, real-instrument, and SPD-pilot extensions, but current build is bounded to simulated practice.
Cost (5/5)	Minimal cost: cardstock, printer, existing camera, and open-source software stack.

The weighted objective analysis confirms the selection of the training-centered path and provides transparent traceability from the decision criteria defined in the objective tree to the final solution selection.

5 Root Cause Analysis

The fishbone head for this analysis is:

Tray content errors during tray reconstruction.

This effect includes missing instruments, wrong instruments, extra instruments, wrong quantities, visible damage or function problems, and item selections that conflict with the count sheet or local tray rule [1]; [3]; [2].

The analysis is organized into six cause categories, each connecting a proximal failure mode to a structural root cause. Three structural drivers appear across categories. First, sterile processing evolved from central supply (materials management) rather than patient care, producing a workforce with weak professionalization and a compensation classification that does not reflect current technical demands. Second, hospital financial accounting treats SPD as a cost center without a revenue line, making every investment visible as expense while error costs remain invisible. Third, no information architecture connects downstream OR outcomes to upstream SPD actions, so errors repeat without triggering systemic correction.

Product.

Same-family instruments (Kelly/Crile, straight/curved Mayo) differ only in subtle dimensional features

with no intrinsic labels or markings [1]; [17]; [3]. Steam autoclaving destroys adhesives and fades inks [18]. Laser etching is expensive and unstandardized across manufacturers, and no regulatory body mandates SPD-level instrument markings [1]; [19]; [20]. Wear and reprocessing cycles further reduce visible distinguishing cues [1]; [17]; [3].

The instrument supply chain evolved around surgical function, not SPD identification. Manufacturers bear labeling cost while SPD receives the benefit but lacks purchasing authority over instrument design [21]. No stakeholder has enough individual incentive to reorganize because error costs are absorbed as diffuse workflow friction rather than concentrated liability [13]. No regulatory body mandates marking because no sentinel event with a clear causal line to patient death has been attributed to unlabeled instruments [22]. Retrofitting across existing inventory is prohibitive.

Tray composition, instrument design, inventory, and unstandardized nomenclature make each tray a complex product to reconstruct reliably [1]. Tray contents are local design decisions rather than fixed universal lists. Procedure-specific, surgeon-specific, and hospital-specific decisions determine which instruments and quantities belong in a set [14]; [23]. A user can know instrument families and still select the wrong item or quantity for a given tray. Reconstruction must also include rejection of visibly unacceptable items, not only selection of named items. Otherwise damaged or suspect instruments can remain in a tray despite correct identity and count [7]; [12].

Knowledge.

Three sub-causes converge. First, knowing the general name is not sufficient. Technicians must see the difference between nearly identical instruments [24]; [3]. Visual discrimination is a perceptual skill that requires deliberate practice with feedback [25]; [26], but the SPD workplace provides no structured opportunity for this practice. Preceptors train on the fly, production pressure prevents deliberate study, and no mechanism identifies which visual discriminations an individual has not yet mastered [3]; [1]. The certification examination failure rates provide evidence that these demands are substantive. HSPA reports CRCST failure rates of 28 to 33 percent, and CBSPD reports CSPDT failure rates near 50 percent [27]. These rates show that SPD work requires genuine technical knowledge of microbiology, sterilization science, regulatory compliance, and advanced instrumentation. The role is classified and compensated as entry-level support labor despite these requirements.

Second, general instrument knowledge and local tray rules are separate domains. Manufacturers produce physically different tools under the same generic name [11]. Tray composition is driven by surgeon preference, hospital case mix, and historical accretion rather than a universal standard [14]; [23]. Local aliases accumulate organically on count sheets with no central nomenclature authority [1]. Count sheets encode tray-specific rules that exist only in that hospital's workflow [11].

Third, certification proves a starting point rather than local readiness. Neither HSPA nor CBSPD mandates recurring competency checks tied to local tray knowledge [27]; [28]; [29]; [30]; [31]. Training inconsistency persists because no single role owns the standard curriculum for each tray. Differences across preceptors, shifts, and training tenures produce conflicting mental models of the same tray [1]; [32]; [33].

Process.

Building a tray forces picking, counting, checking, and arranging into one step with no double-check system. Alfred et al. found that 55.0 percent of tray defects occur during assembly [1]. Nichol et al. found that 88.6 percent of observed errors involve visualization tasks such as inspection, identification, function checking, and sorting [3]. Sequential verification would double labor per tray, and no accreditation mandate from the Joint Commission, CMS, or HSPA requires a two-person check for tray assembly [1]; [34].

Productivity metrics structurally incentivize single-pass assembly. The AAMI benchmarking framework

uses worked-hours-per-unit (WHPU), reinforcing throughput as the primary performance signal [35]. Instrument damage such as corrosion, stains, burrs, and worn jaw serrations can be missed when inspection relies on rapid visual checks under poor lighting [17]. Workers use only their eyes and hands, without magnification, side-by-side comparison guides, or decision-support tools at the workstation [3]. No worker can remember every item in every tray when a single hospital may manage hundreds of tray configurations, each with 30 to more than 100 instruments. Production pressure and cognitive load cause workers to break from the count sheet and rely on memory, especially for familiar trays [11]. No formal measurement of count-sheet compliance exists, so the gap between assumed behavior and actual behavior is invisible to management. Tray errors have not crossed the sentinel-event threshold needed to trigger regulation [22].

Information.

Count sheets become incomplete or outdated because tray contents change continuously. Instruments are added, removed, substituted, or replaced as surgeon preferences shift, manufacturers discontinue items, or vendors supply new loaners. No dedicated role or budget exists for count-sheet maintenance. Updating a count sheet is a manual, discretionary task that competes with production work. Without a formal change-management workflow, sheets accumulate inaccuracies across multiple revision cycles [11]. Missing or outdated identification information, IFUs, and tray listings at the workstation force workers to rely on memory [36].

Loaner trays introduce a separate information failure. They arrive with manufacturer-stocked contents and vendor documentation that uses the vendor’s naming conventions and part numbers, not the hospital’s local aliases or count-sheet format. SPD receives the tray with hours or minutes to process it before surgery, and the worker must reconcile the vendor’s nomenclature against the local count sheet with no lead time to prepare or cross-reference [37].

Environment.

Multiple environmental conditions reduce the attention and time available for tray reconstruction. Pressure and interruptions break focus for detail-heavy visual work [1]; [38]. The open-plan workstation layout makes it difficult to shield assembly workers from interruptions. Phones ring with OR requests, surgeons call for add-on instruments, and runners deliver and collect trays [38]. Uncomfortable workstations, awkward reaches, and poor setup make a job that already needs close attention harder [39]. Ergonomic improvements compete for facilities budget with clinical areas that have more organizational visibility. SPD staff have less organizational voice than clinical departments [40].

The wage level is a structural cause that amplifies every other category. SPD is a cost center within the hospital. It generates no direct revenue, so staffing budgets are structurally constrained relative to OR capacity, which is a revenue center [41]; [42]. Hospitals routinely add surgery suites without proportionally increasing SPD headcount [43]. Average SPD pay is \$69,217, with a 14 percent drop in respondents receiving raises. SPD workers describe wages as “less than your local Starbucks wage” [44]; [45]. Fewer staff means less time for training and less thorough checking [1]; [29]; [46]; [47].

Three structural mechanisms produce this wage level. First, sterile processing evolved from central supply (materials management) rather than patient care. The role retains a compensation classification from that origin, despite technical demands that now include microbiology, sterilization science, and regulatory compliance. Hospital associations resisted mandatory certification in the 2000s in part because certification would create upward wage pressure [48]. Second, the absence of professional licensure means that no regulatory body establishes a minimum compensation baseline. Unlike nursing or surgical technology, SPD has no individual license, no board with enforcement authority, and no mandated competency infrastructure. The travel market premium of 40 to 80 percent over permanent staff pay reveals the market-clearing wage that the permanent classification system suppresses [49]. Third, low union representation means no institutional mechanism exists to counter the cost-center budget logic.

Healthcare technician unionization remains below the 13.2 percent healthcare average, falling as low as 5 percent in right-to-work states [50]; [51]. SPD's physical isolation from clinical staff makes organizing difficult, and most healthcare union drives target nurses around staffing ratios rather than technician compensation [52]. Without collective bargaining, the self-reinforcing cycle continues: structural underinvestment produces low wages, low wages produce chronic vacancies, vacancies increase production pressure, and production pressure preempts the training and process improvements that could reduce errors.

Policy & Feedback.

SPD governance is decentralized across shifts. No single role is responsible for maintaining, auditing, or enforcing count-sheet accuracy or tray-assembly procedure across all shifts [11]; [53]. Creating a single authoritative standard for each tray requires a designated owner, a maintenance process, and enforcement across shifts, but none of these are budgeted.

Certification bodies (HSPA, CBSPD) recommend but do not mandate recurring competency checks tied to local tray knowledge [30]; [28]. When competency programs exist, they are designed and administered locally with no external audit [54]. The absence of professional licensure (described in the Environment section) means that no external body holds hospitals accountable for maintaining a minimum competency standard.

Error reporting is manual, slow, and deprioritized. Errors discovered during surgery are documented, if at all, in incident reports that may take weeks to reach SPD management. By that time the specific tray, technician shift, and training gap are hard to identify, so the error teaches no preventive lesson [3]; [2]. The hospital incident reporting system was designed for clinical errors such as medication errors, falls, and wrong-site surgery. Tray-content errors do not fit neatly into the reporting categories, so they are underreported. Without closed-loop tracking that ties an error back to a root cause, downstream detection does not drive upstream improvement [3]; [2].

Consolidated Root Cause Chain.

The six categories above share three structural drivers. Together they explain why the same errors recur across hospitals and over time despite being well documented.

Sterile processing evolved from central supply (materials management) rather than patient care. That historical origin produced a workforce with weak professionalization: no individual licensure, no board with enforcement authority, and no mandated competency infrastructure. The role carries a compensation classification from its materials-management origin that does not reflect current technical demands. Hospital associations resisted mandatory certification in part because certification would create upward wage pressure. Low union representation leaves no institutional mechanism to counter the cost-center budget logic that keeps wages below market-clearing levels. The travel market premium of 40 to 80 percent over permanent staff pay reveals the suppressed wage.

Hospital financial structure separates revenue generation (OR) from support cost (SPD). Every SPD investment is visible as expense, while the benefit of fewer errors is invisible in standard metrics. Hospitals add surgery suites without proportionally increasing SPD headcount. The same financial structure makes it rational for each individual hospital to underinvest in SPD wages and training because the cost of errors is absorbed as diffuse workflow friction rather than direct liability.

No information architecture connects downstream OR outcomes to upstream SPD actions. The feedback loop from error to correction is broken, so the same failures repeat without triggering systemic change. Error reporting is designed for clinical events, not tray-content errors.

The system remains stable because no stakeholder has both the incentive and the authority to change it. Manufacturers control instrument design but bear no cost from SPD identification errors. Hospital administrators control SPD budgets but see wage increases as expense and error costs as invisible friction.

SPD workers perform the work but lack the union representation, professional licensure, or organizational voice to demand structural change. Regulatory bodies have not acted because tray errors have not crossed the sentinel-event threshold.

Final Root Cause Framing.

Tray content errors during reconstruction are a system problem caused by the interaction of local tray complexity, fine-grained visual identification, count-sheet information quality, hands-on local knowledge, manual process controls, workforce conditions, and weak feedback loops. The workforce conditions that amplify every other category have structural origins: the historical classification of SPD as materials management, the absence of professional licensure and union representation, and a cost-center accounting model that makes wage investment visible as expense while error costs remain invisible.

Short version:

Tray reconstruction is vulnerable because local tray rules must be translated into accurate visual identification, counting, rejection of distractors, and damage recognition during a manual assembly task. Workforce conditions determined by historical classification, weak professionalization, and low union representation control how much time and attention are available for that translation.

The strongest evidence points to local instrument knowledge, count-sheet use, visual identification, assembly defects, competency validation, workforce conditions, and feedback as testable contributors that can be represented in a simulated tray task [1]; [3]; [11]; [28]; [46].

6 First Attempt: A Computer-Vision Tray Checker

6.1 CV Literature Context

Computer vision remains technically plausible, but it is no longer the central project proof. Deol et al. show that automated surgical-instrument detection and counting can work in an experimental proof-of-concept setting [55]. Atabuzzaman et al. show that structured image acquisition can support ultra-fine-grained surgical instrument classification, and Xin et al. address CSSD-oriented instrument recognition [56]; [57]. These papers support CV as authoring support, camera-assisted practice, or a future visual review layer, but real SPD deployment would need local data, manufacturer and variant coverage, lighting and glare testing, occlusion handling, open-set behavior, workflow review, human confirmation, false-alert analysis, cleaning and device-policy review, and site-specific failure-mode reporting. Kienle et al. show why this boundary matters: strong in-domain detection can drop when instruments come from different manufacturers [58]. The project’s earlier CV pipeline is still useful technical evidence, but it should be treated as a support layer. The project’s primary claim should come from the training loop because that loop can be built, tested, and interpreted without pretending that deployment-grade instrument recognition has already been solved.

6.2 Requirements for Automated Verification

The tray-status logic used a two-threshold confidence band with a scan-quality gate, because a single detector threshold is insufficient to distinguish a truly absent item from one missed due to glare, occlusion, blur, similar-class confusion, or open-set behavior:

Table 7: Original planned tray-status logic using two-threshold confidence band

User-Facing State	Decision Logic
Present	Required class has at least required count of detections above high threshold T_present , no active similar-class or image-quality risk flag, and duplicate boxes reconciled.
Needs Review	Detections between T_review and T_present , similar class appears, count unstable, unknown or extra object plausible, or scan quality degraded.
Missing	No candidate above low threshold T_review , and scan quality acceptable enough that absence is meaningful.
Rescan Recommended	System lacks evidence but glare, occlusion, blur, poor framing, or lighting makes a missing label unsafe.

The technician-facing UI was designed around the tray decision with one primary action per state, visual-first feedback, alert restraint, and explicit human sign-off:

Table 8: Original planned technician UI requirements

Requirement	Design Direction
One primary action per state	Show the next meaningful action: start scan, rescan, review issue, confirm tray, or stop.
Clear tray-status hierarchy	Summarize Complete, Missing, Extra, and Needs Review above detailed detections.
Fast correction and rescan	Provide direct correction, dismiss, and rescan paths instead of forcing a workflow restart.
Specific review reasons	Label review prompts with the cause: low confidence, similar-class risk, glare, overlap, unknown object, or count mismatch.
Visual-first feedback with optional sound	Persistent visible status as the source of truth; short optional audio for major state changes.
Alert restraint	Escalate only conditions that change the tray decision or require user action.
Human sign-off	Require explicit final confirmation. The system supports the decision, not silently makes it.
Manual fallback	Make it clear how to continue if camera, model, or scan quality is unavailable.

The original system also included a physical capture fixture: an overhead camera stand mounted on a portable weighted board, designed for tabletop use with an optional rolling standing base. The board-mounted design preserved repeatable camera geometry across work surfaces without requiring table modification.

The original system also included a physical capture fixture: an overhead camera stand mounted on a portable weighted board, designed for tabletop use with an optional rolling standing base. The board-mounted design preserved repeatable camera geometry across work surfaces without requiring table modification.

6.3 System Design: Camera-Supported Tray Check

The original product concept was a tray-verification tool that used a camera, a YOLO object detector, and confidence-based logic to check whether a technician’s assembled tray matched the required instrument list. The system was designed to support the following workflow:

1. Load or enter the required instrument list for a tray.
2. Capture the current tray view with a camera.
3. Detect visible instruments and assign class labels with confidence scores.
4. Aggregate detections into observed class counts.
5. Compare observed counts against required counts.
6. Present a result that highlights what appears present, missing, extra, or uncertain.
7. Preserve the final decision as human judgment rather than silent automation.

The tray-status logic, technician-facing UI requirements, and physical capture fixture design are documented in Section 6.2 above. The System Design section below describes the final training-centered system that replaced this CV-based approach.

6.4 CV Experiments and Results

The CV experiments tested how detection degrades when training and test conditions differ along three axes: illumination, surface reflectivity, and layout occlusion. All staged experiments used YOLO11s on a controlled self-collected dataset with four instrument-like classes (spoon variants) arranged in a fixed 11-step brightness ladder (800 to 80 estimated lumens) across two background conditions (matte, reflective) and two layout conditions (separated, overlapping). The design isolated each variable independently rather than mixing effects.

Lavado baseline. A YOLO11s model was also trained on the Lavado surgical-instrument dataset: 2,106 training images across four real instrument classes (scalpel, straight dissection clamp, straight Mayo scissors, curved Mayo scissors). After 100 epochs the model achieved precision of 0.819, recall of 0.965, mAP50 of 0.990, and mAP50-95 of 0.944 on the held-out test set (302 images). This established that the training pipeline and architecture perform competitively on a published surgical-instrument dataset before the staged proxy experiments.

Brightness degradation. Two experiments measured monotonic mAP50-95 decay as the train–test illumination gap widened. `brightest_train_darker_test` trained on 800-lumen separated images and tested across all dimmer levels, producing mAP50-95 of 0.844. `darkest_train_brighter_test` trained on 80-lumen separated images and tested across all brighter levels, producing mAP50-95 of 0.781. The decay was asymmetric: the brightest-trained model retained higher recall under moderate drops, while the darkest-trained model recovered more quickly as light increased. Per-brightness metrics confirmed the trend was consistent across both matte and reflective backgrounds.

Background transfer. Two experiments tested whether the model learned object shape or overfit to the scene background. `matte_train_reflective_test` trained on matte-background images and tested on reflective-background images across all brightness levels, producing mAP50-95 of 0.759. `reflective_train_matte_test` trained on reflective-background images and tested on matte-background images, producing mAP50-95 of 0.830. The reflective-to-matte direction was stronger, suggesting that reflective training data provides broader feature invariance.

Occlusion stress. The separated-to-overlay layout shift produced the largest degradation. `separated_train_overlay` trained on non-overlapping layouts and tested on overlapping layouts, producing mAP50-95 of 0.389 compared with 0.995 on the matched-condition baseline. This gap was consistent across all brightness ranks and both backgrounds, showing that training on neatly arranged instruments does not transfer to cluttered arrangements. The overlay condition was the only experiment where recall fell below 0.70, making it the clearest product boundary: the system should request a rescan whenever tools overlap.

Table 9: Aggregate test-set metrics for the seven staged brightness experiments

Stage	Precision	Recall	mAP50	mAP50-95
<code>brightest_train_darker_test</code>	0.840	0.969	0.991	0.844
<code>darkest_train_brighter_test</code>	0.819	0.969	0.972	0.781
<code>bright_train_dim_test</code>	0.996	1.000	0.995	0.984
<code>dim_train_bright_test</code>	0.996	1.000	0.995	0.979
<code>matte_train_reflective_test</code>	0.931	0.900	0.929	0.759
<code>reflective_train_matte_test</code>	0.834	0.900	0.968	0.830
<code>separated_train_overlay_test</code> (overlay)	0.709	0.660	0.627	0.389
<code>separated_train_overlay_test</code> (baseline)	0.998	1.000	0.995	0.995

Lighting diversity (spoon proxy). A separate experiment confirmed that adding lighting variety during training improves held-out-condition detection. Three YOLO11s models were trained on the same Lavado instrument classes with increasing lighting diversity (reference lighting only, reference plus 45-degree flashlight, reference plus 45- and 90-degree flashlight). Each additional lighting condition improved precision and recall on held-out test conditions, with the most diverse training set producing the highest and most stable metrics across all held-out lighting angles.

What the cumulative evidence meant for the project. Across all experiments the pattern was consistent: detection worked reliably in controlled, matched-condition settings and degraded under every variation that SPD workflows present—variable illumination, reflective tray surfaces, and overlapping instruments. The occlusion condition (mAP50-95 = 0.389) was the clearest operational boundary: the system can support technician review when tools are separated, but must request a rescan or flag uncertainty when overlap is detected. These results showed that controlled lighting and layout are prerequisites for reliable detection, and that real SPD workstations would require site-specific training data to cover the range of conditions encountered. The pivot to a training-centered platform preserved the data collection pipeline, camera code, and detection experiments as foundation for future camera-assisted practice. The primary evidence path shifted from CV generalization to measurable novice learning through a simulated tray task with physical AprilTag cards, a claim that could be supported with the project’s available resources.

6.5 Boundary Conditions for Future CV Work

TODO: Summarize the conditions under which a CV-based tray checker would be viable: (1) access to site-specific training data covering local instruments, lighting, and layouts; (2) a closed-set assumption or robust open-set rejection; (3) human-in-the-loop confirmation for low-confidence and lookalike detections; (4) per-manufacturer validation (Kienle et al. 2025); (5) workflow integration testing in a live SPD environment.

7 Literature And Related Work

The root cause analysis identified several interconnected causes of tray-content errors. Expanding each cause to its structural driver reveals four common threads across all six categories. First, sterile processing evolved from central supply (materials management) rather than patient care, producing a workforce with weak professionalization. Second, hospital financial structure separates revenue generation from support cost. Third, no information architecture connects downstream OR outcomes to upstream SPD actions. Fourth, the SPD workforce lacks the organizational voice that clinical departments have. Together these four drivers explain why the individual root causes persist despite being well documented.

7.1 The Solution Landscape

The solution space for reducing tray-content errors during reconstruction can be organized into seven categories by intervention point. Each category targets a different subset of the root causes, and each faces a different feasibility constraint for this project.

The seven identified solution categories are:

1. **Automated content verification:** post-assembly inspection using CV, RFID, or barcode scanning to verify tray contents.
2. **Physical error-proofing:** modifying the tray or instrument interface (custom foam, color coding) to prevent selection errors.
3. **Workflow / process design:** restructuring task handoffs and verification steps through dual checks or gating sign-offs.
4. **Decision support at point of assembly:** providing standardized count sheets, photos, and alias lookup at the workstation.
5. **Task simplification:** rationalizing tray contents and standardizing instrument sets across procedures.
6. **Workforce / accountability systems:** competency reassessment, closed-loop error tracking, and structured mentorship.
7. **Training and simulated practice:** building local tray skills through retrieval practice, simulation, and feedback before supervised work.

Table 10: Seven solution categories compared by constraint

Category	Intervention point	Example solutions	Root causes addressed	Key constraint for this project
Automated content verification	Post-assembly inspection	CV tray checker, RFID tray scan, barcode line scan, weight check	Product: wear hides differences and Process: no backup check exist	Requires real instruments, lighting, workflow integration; cannot validate without SPD deployment
Physical error-proofing	The tray or instrument interface	Custom foam cutouts, color-coded handles, segregated tray zones	Knowledge: workers must memorize tiny visual details and Process: assembly lacks a separation of pick from check	Expensive custom fabrication per tray; difficult to validate and maintain across tray variants
Workflow / process design	Task structure and handoffs	Dual verification, gating sign-offs, pre-sorted pick lists	Process: no structured backup check during assembly and Policy: nobody formally owns the standard	Primarily organizational change; no technical prototype to build or measure in a controlled study
Decision support at point of assembly	Information available during re-construction	Standardized count sheets with photos, alias lookup, reference kiosk	Information: outdated count sheets force guessing and Knowledge: general instrument names do not convey local tray rules	Improves the reference but does not exercise or measure the technician's own ability
Task simplification	The tray definition itself	Tray rationalization, instrument consolidation, procedure-level standardization	Knowledge: differentiating lookalikes requires local familiarity and Environment: pressure and interruptions break focus	Requires hospital utilization data and multi-stakeholder buy-in; no buildable prototype
Workforce / accountability systems	Competency management and error feedback	Periodic competency reassessment, closed-loop error tracking, structured mentorship	Policy: standards are only tested at hiring and Feedback: slow error reporting lets mistakes repeat	Primarily organizational policy change; no measurable prototype within our project scope
Training and simulated practice	Pre-work and between-work skill building	VR simulation, digital flashcards, tangible tray simulation, structured OJT	Knowledge: lookalike discrimination requires repeated practice and Local knowledge: tray-specific names and rules are not taught in general curricula	Requires learning-science justification and a novice study, both accessible without SPD deployment

Each category reduces a real risk. Categories 1 through 6, however, all depend on resources that this

project cannot access within our project timeline: SPD deployment sites, real instrument inventories, hospital utilization data, institutional policy authority, or multi-stakeholder buy-in. Category 7 — training and simulated practice — is the only category whose primary evidence can be collected with accessible participants (novices), low-cost materials (printed cards), and a bounded pre/post study design on a simulated local tray task. Section 7 examines the literature supporting each relevant area in depth.

7.2 SPD Error and Tray-Readiness Evidence

7.3 SPD Training and Competency Gap

The RCA (Section 5) identified the gap between general certification and local tray competence: HSPA’s CRCST requires 400 hours of hands-on experience, but certification alone does not cover local instrument names, tray rules, or sustained lookalike discrimination [27]. CBSPD’s CSPDT route similarly treats certification as minimum competency, allowing candidates to qualify through work experience, allied health plus SPD experience, a sterile processing course, or related product sales or service experience [59].

SPD training in practice follows a blended pattern that combines coursework, hands-on practice, and preceptor-led onboarding:

Table 11: SPD training patterns and TrayGuard implications

Training pattern	Evidence	Implication for TrayGuard
Certification exam plus verified hands-on hours	HSPA CRCST requires 400 SPD hands-on hours and a computer-based exam; CBSPD CSPDT uses eligibility routes that include work experience or a course plus exam [27]; [59].	The tool should measure learning and support practice before and during supervised competency development.
Self-paced or classroom instruction	Towson’s ed2go-backed course is self-paced online, 190 hours, covering instrument identification and tray assembly. Altamont’s program is 135 hours over 8-12 weeks with live online classes, in-person labs, and an optional 240-400 hour externship [60]; [61].	The platform can complement formal instruction with reusable local modules and repeated practice.
Preceptor-led onboarding	HPN describes hospital examples including 90-day on-the-job training programs, 19-week programs with textbook chapters, quizzes, and progress tests, and structured 1:1 preceptor onboarding with presentations, online modules, written tests, direct observation, and one-on-one training [31]. HSPA training guidance frames core program elements as structured orientation, competency documentation, and ongoing education; training that stops after orientation perpetuates inconsistent habits [62].	The system should reduce low-level explanation burden and give preceptors evidence of where a learner still struggles.
Simulation and skills assessment	AORN describes Penn Medicine’s earn-to-learn SPD program combining didactic education, hands-on training, simulation, mentorship, content testing, and skills-based assessment [29].	Simulated tray sorting and pre/post tasks are aligned with existing training practice.

The common thread across patterns is that hands-on practice and preceptor attention are the scarce resources. Several program examples show the gap between classroom preparation and clinical capacity. Penn Foster’s online sterile processing program prepares learners for the CRCST exam but does not require the exam or the 400 hands-on hours [63]. MedCerts states that full CRCST certification requires 400 hours of hands-on experience following provisional certification [64]. AIMS Education requires a 400-hour clinical internship, but its schedules depend on clinical-site availability [65]. Central Sterilization Solutions states its classroom course has no hands-on training in the classroom and that hands-on training is conducted during externship [66]. These examples illustrate the training-access gap: repeatable, feedback-rich practice with local instruments and tray rules is hard to scale before and during real SPD experience because it depends on site access, clinical scheduling, preceptor attention, and available teaching materials.

Broader allied-health placement research supports treating clinical placements as capacity-constrained environments. A Queensland study found that placement-offer growth did not keep pace with student-program growth, and barriers centered on resourcing and increasing demand [67]. A 2025 Greater Sacramento report identified shortage of clinical placement opportunities as a barrier to workforce pipeline growth, and highlighted student preparedness as part of successful placement partnerships [68]. TrayGuard gives learners repeated practice before, during, or between supervised work, and gives educators evidence about weak instruments, weak tray rules, high-confidence errors, and uncertainty. This addresses the practical constraint: supervised training time is a scarce resource [29]; [47]; [46].

7.4 Local Tray and Count-Sheet Evidence

The RCA (Section 5) identified incomplete count sheets and local tray variation as root causes. Count sheets define tray names, sections, contents, quantities, reference numbers, placement, and sign-off fields (Nadeau, 2024; Stryker, 2023). Tray contents are design choices shaped by procedure, surgeon, and hospital policy (dos Santos et al., 2021; Ahmadi et al., 2023), so general instrument knowledge does not transfer across sites. Loaner trays add further complexity through scheduling, delivery, IFU, and count-sheet dependencies (STERIS, 2021). TrayGuard stores local names, aliases, approved photos, required counts, lookalike pairs, distractors, assessment variants, and module version. Modules must be instructor-verified before results are described as content-valid.

FGVC-derived modules serve a narrower prototype role. Atabuzzaman et al. collected ultra-fine-grained surgical instrument images from major, minor, and eye tray contexts [56]. TrayGuard’s current FGVC prototype uses the major-tray image archive and class-name mapping to create image cards and same-family discrimination exercises. The FGVC source provides observed instrument classes and representative images; public count sheets provide the field structure for tray names, sections, quantities, locations, check boxes, and sign-off fields [69]; [70]. The prototype’s FGVC quantities are simulated as one card per selected class, and exact local tray membership remains an instructor-review variable.

7.5 Learning Science and Simulation Evidence

Health professions education provides a general precedent for simulation: a broad review of technology-enhanced simulation found improved knowledge, skills, and behaviors compared with no intervention, and simulation-based medical education with deliberate practice has also been shown to outperform traditional clinical education in meta-analysis [71]; [72]. Earlier medical simulation guidance identifies feedback, repetitive practice, curriculum integration, and measurable outcomes as important features of effective simulation [73]. For sterile processing specifically, Ofstead et al. provide the closest direct precedent. Their borescope/endoscope visual-inspection training pilot used a pre-test, structured teaching, hands-on practice, image-based testing, confidence/satisfaction measures, workplace homework, and a delayed booster with certified sterile-processing professionals [28]. Tray assembly is a different task, but the learning structure maps well: the learner must build visual familiarity, recognize subtle dif-

ferences, decide when uncertain, apply a procedural rule, and improve between a baseline task and a later task. Retrieval-practice evidence explains why the prototype should ask learners to produce answers before showing them. Tests can improve long-term retention relative to restudy, repeated testing improved long-term retention in a medical-education randomized trial, and reviews support practice testing, distributed practice, spaced digital education, electronic flashcards, and actionable feedback in health-professions learning [74]; [75]; [76]; [77]; [78]; [79]. This supports retrieval-first cards, quizzes, weak-item repetition, and error-specific feedback as the instructional core rather than decoration around the tray task. The electronic-flashcard evidence is especially relevant to the proposed study and card modes. Barrison et al.’s 2025 scoping review found that electronic flashcards are widely used in health-professions education, that the research base grew rapidly from 2019 to 2024, and that many studies examine utilization and associated learning outcomes. The review also cautions that development and delivery methods are less systematically studied [78]. For TrayGuard, that means flashcards are a credible medical-learning mechanism, but they should be designed as one part of a larger tray simulation: short prompts, single target concepts, local photos or card representations, immediate answer feedback, spacing or weak-item repetition, and a downstream tray-sort task that tests application rather than isolated recall. The proposed AprilTag-card tray is a reasonable physical proxy because it preserves the target cognitive actions while replacing regulated instruments with safe, cheap, repeatable tokens. The user still has to read a local tray list, retrieve which items belong, distinguish distractors and lookalikes, apply counts, place items into a tray area, and respond to feedback. The marker makes the simulated choice observable to the app. NeuroVase is a close emerging design precedent: it uses tangible cue cards with a tablet-based mobile AR system, structured medical curriculum, pre/post assessment, usability measures, and a controlled user study for neurovascular anatomy and stroke education [80]. QR-code education literature similarly supports low-cost printable codes as an access layer for healthcare learning, simulation, and training support [81]. The same design pattern appears in workplace training research. Meta-analytic evidence shows that organizational training improves learning, behavior, and operational results when it includes practice and feedback [82], and Salas et al. argue that effective training should be treated as a system: analyze the work, design instruction around target competencies, provide practice and feedback, and evaluate transfer [83]. TrayGuard follows that model by targeting a specific work task (tray reconstruction), designing instruction around instrument identity and count-sheet rules, providing repeated practice with error-specific feedback, and measuring pre/post transfer to a comparable tray variant. Serious tabletop and board-game precedents also support the choice to turn a healthcare protocol into a bounded physical learning game. Ward et al. designed and evaluated the PlayDecide patient-safety board game in two acute teaching hospitals; the intervention used cards and facilitated discussion to teach junior doctors about safety concerns and reporting, and the authors concluded that it was valuable for patient-safety education and deep discussion [84]. Their study demonstrates that abstracted physical artifacts can stand in for clinical objects when the study measures learning, discussion, decision quality, and protocol use. Marker cards are the first prototype for measuring novice improvement on simulated local tray sorting. A follow-on SPD pilot would replace or supplement cards with local photos, real teaching instruments, educator-reviewed count sheets, and supervised workplace validation.

7.6 Market and Comparable Tools

Comparable systems cluster into four groups: tray-management products, instrument tracking systems, surgical simulation tools, and emerging CV or robotic research systems. None of them eliminates the need for TrayGuard’s bounded training claim.

Table 12: Comparable tools and gap relative to TrayGuard

Tool or approach	Target user	Relevant features	Evidence strength	Gap relative to TrayGuard
Tray Pacer	SPD teams and managers	Count sheets, tray assembly support, photos, productivity and proficiency metrics.	Commercial product evidence.	Supports operations, but public materials do not establish the specific pre/post local tray learning study proposed here [85].
LayerJot SID	SPD and surgical instrument teams	Instrument documentation, photos, count sheets, and tray information.	Commercial product evidence.	Strong local-information precedent, but not a demonstrated tag-card novice learning loop [86].
CensiTrac and similar tracking systems	Hospitals, SPD, OR inventory teams	Instrument and tray tracking, traceability, inventory visibility.	Commercial and operational precedent.	Addresses tracking and lifecycle visibility more than low-cost novice simulation and weak-item training [87].
Touch Surgery and surgical simulators	Medical trainees and clinicians	Repeated simulation attempts, performance logging, skill practice.	Published evaluation and commercial adoption.	Supports simulation logic, but focuses on surgical procedure skills rather than SPD tray reconstruction [88]; [89].
NeuroVase-style tangible cards	Health-professions learners	Physical cue cards, mobile AR, structured curriculum, pre/post testing, usability measures.	Emerging preprint evidence.	Strong design precedent for tangible medical learning, but not sterile-processing content [80].
Surgical-instrument CV research	Researchers and future technical teams	Detection, counting, fine-grained classification, instrument-stand recognition.	Published research, usually bounded datasets.	Supports future camera assistance, but not deployment-ready SPD workflow claims [55]; [56]; [58].
Autonomous tray assembly research	Robotics researchers and future automation teams	Manipulation, sorting, and assembly concepts.	Early research.	Much higher hardware, sterility, safety, and integration burden than the project's training loop [90].

TrayGuard’s contribution is a low-cost local training loop that turns a tray module into retrieval-first study, applied tray sorting, immediate feedback, no-hints assessment, and exportable pre/post evidence.

Medical training software already uses the same design pattern of simulation, guided practice, feedback, scoring, and progress tracking. Touch Surgery provides mobile surgical procedure simulations with repeated attempts and scoring, and a validation study found that users improved across repeated attempts while higher-experience cohorts outperformed lower-experience cohorts [88]. Osso VR demonstrated in a randomized trial that VR-trained novices completed a downstream orthopedic procedure more often, made fewer incorrect steps, and finished faster than guide-only learners [91]. Body Interact uses screen-based virtual patient scenarios with immediate feedback, auto-grading, and educator analytics [92]. Surgical Science and Simbionix provide commercial medical simulators across specialties, positioning simulation-based assessment as a credible healthcare training category [93]. These precedents support treating TrayGuard as a health-professions simulation tool that applies established learning mechanisms to a new domain: local SPD tray reconstruction.

7.7 Literature Gaps and System Response

The literature supports each design choice individually, but no existing work closes the specific gap that TrayGuard targets: a low-cost, repeatable, locally configurable training loop for SPD tray reconstruction that measures pre/post simulated improvement.

Table 13: Literature gaps and TrayGuard’s system response

Gap in the literature	System response	Stakeholder meaning
CV papers show instrument-detection feasibility while leaving manufacturer, site, tray, and workflow variation thinly tested [55]; [56]; [58].	Keep CV optional and locally verifiable. Use instructor-approved content as the source of truth for the training evidence.	The tool does not depend on deployment-grade CV. Camera support can be added later after site-specific validation.
SPD error papers document real defect rates while leaving the training-intervention mechanism unresolved [1]; [3].	Run a pre/post simulated learning study with error-category breakdown.	Students can test novice learnability; SPD trainees remain the future validation group for workplace transfer.
Training literature supports retrieval practice, simulation, spacing, and feedback while leaving the SPD tray-module design open [74]; [71]; [28].	Build the module around local retrieval cards, quizzes, applied sorting, weak-item repetition, and error-specific feedback.	Trainees get repeatable practice; instructors get evidence about weak items.
Adoption literature provides broad healthcare-AI guidance without SPD training-assistant specifics [94]; [95].	Report workflow friction, role boundaries, privacy assumptions, and pilot requirements alongside the prototype.	Administrators get a clearer approval path and an explicit future-evidence list.

8 Final Product Concept And Design Rationale

8.1 Final Product Concept

TrayGuard is a local tray training and assessment platform. The core product is a learner-facing app plus a file-backed tray module that defines the instruments, aliases, required counts, distractors, lookalike pairs, distinguishing features, and assessment variants for one local tray. The first build should use printable AprilTag instrument cards as the physical simulation layer. Each card represents one instrument or instrument variant. The app scans the cards on a tabletop or tray area, treats the detected markers as the learner’s tray selection, scores the result against the tray template, and returns missing, extra, wrong, misidentified, and wrong-count feedback in practice mode. Pre-test and post-test modes suppress hints and corrective feedback so the same system can collect baseline and post-training evidence. The full learning loop is:

```
local tray module -> pre-test -> retrieval-first cards -> quiz -> AprilTag-card tray sort -> post-test -> metrics export
```

This concept keeps the prototype buildable while preserving the important task structure: the learner must use local names, local counts, visual/semantic distinctions, and tray rules to assemble a simulated tray.

8.2 Rationale for the Training-Centered Pivot

The project originally aimed at a live CV tray checker: detect instruments, compare detections against a tray list, and return a readiness result. That direction remains attractive, but it has the wrong proof

burden for our current project. A useful clinical tray checker would need strong evidence across real instruments, local tray definitions, similar variants, glare, occlusion, unknown objects, workflow interruptions, false alerts, human review, device policy, cleaning, privacy, and adoption. The current project can demonstrate parts of that pipeline, but it cannot responsibly claim deployment-grade clinical reliability. The Week 6 pivot made the value proposition more testable: reduce simulated time-to-competency and training burden for sterile-processing learners by building an interactive training platform. That direction keeps the same root problem, because missing, wrong, extra, and misidentified instruments still arise during tray reconstruction. It changes the claim from "the model can outperform humans in a hospital environment" to "novices can improve on a measured local tray simulation after retrieval-centered practice." This is a better project claim for three reasons. First, it matches the evidence: sterile-processing work depends on local knowledge and supervised practice, and simulation/retrieval evidence supports repeated practice with feedback. Second, it is buildable with low-cost materials: a file-backed tray module, printable cards, a scanning surface, scoring, and export. Third, it reduces risk for a future CV system by creating validated local content, error categories, workflow language, and learner data before asking a camera model to make clinical-grade distinctions.

8.3 Alternatives Considered

Several product directions were considered:

Table 14: Alternative product directions considered and rejected

Alternative	Benefit	Reason rejected or deferred
Deployment-grade CV tray checker on real instruments	Closest to the original automation concept and strongest if validated in an SPD.	Requires local instrument data, clinical workflow integration, failure-mode validation, and regulatory/adoption work beyond our project timeline.
Software-only study and quiz module	Fastest implementation path and easiest novice study.	Too far from the tray reconstruction task; it would mostly test recognition, not applied count-sheet use.
Real teaching instruments with camera-supported tray review	Better physical fidelity than cards.	Requires access to representative instruments, cleaning/handling rules, storage, and educator review; appropriate for an SPD pilot after the simulated loop works.
Printable QR/AprilTag instrument cards	Cheap, portable, repeatable, and compatible with device scanning; closely matches NeuroVase-style tangible cue-card learning.	Less physically realistic than real instruments, so claims must stay limited to simulated tray-task improvement.
Manual drag-and-drop tray simulation	Simple software implementation without camera setup.	Does not exercise physical search, placement, or scanning workflow, and is less persuasive for an interactive project demo.

The selected first prototype is the printable AprilTag-card path because it creates a physical, measurable simulation without depending on deployment-grade instrument recognition.

8.4 Selected Candidate and Backup Paths

The selected candidate is:

A local tray training and assessment app that uses printable AprilTag instrument cards for simulated tray sorting, retrieval-first study, quiz, practice feedback, no-hints pre/post assessment, and metrics export.

The backup paths are layered so the project still produces evidence if one piece fails.

Table 15: Backup path strategy by risk level

Level	Selected path	Backup path	Claim preserved
Learning feature	Retrieval cards, quiz, practice sort, pre/post sort.	If the full loop is too long, run pre-test, study cards, practice sort, and post-test only.	Immediate simulated learning.
Physical interface	AprilTag cards scanned on a tray mat.	Use QR codes or manual card selection if AprilTag scanning is unreliable.	Local tray-task learning, with marker reliability reported as a limitation.
Content	basic_general_tray_v1 with 8-12 required concepts, distractors, lookalikes, and matched variants.	Reduce to 6-8 required concepts while preserving at least two lookalike pairs and count errors.	Bounded module feasibility.
CV	Marker detection only for project scoring.	Defer all camera recognition and score user selections manually.	Training evidence without CV overclaim.
Evaluation	8-12 novice participants plus delayed retention if schedule allows.	Run a smaller formative walkthrough and report it as usability/protocol evidence, not learning proof.	Risk-reduction handoff.
Content validity	SPD educator or instructor review.	Faculty or surgical-technology reviewer plus explicit future SPD review requirement.	Plausibility review, not clinical validation.

This backup strategy is intentionally conservative. It protects the project from collapsing into a fragile detector demo and keeps the main deliverable aligned with the training pivot.

8.5 Rationale for AprilTag Marker Choice

The prototype uses printable AprilTag fiducial markers on physical instrument cards. This section justifies AprilTag specifically against the alternatives of no physical component, QR codes, ArUco markers, and real instruments with camera-based recognition.

8.5.1 Why Physical Markers at All

A purely digital drag-and-drop tray simulation would be simpler to implement and would avoid camera-setup and detection-reliability concerns. It is rejected because the target task is fundamentally physical: SPD technicians search for, pick up, examine, count, and arrange real instruments under a tray boundary [3]; [1]. A purely on-screen interface removes the spatial search, the need to distinguish similar objects

in hand, and the physical count-and-place workflow that the study is meant to simulate. Physical cards preserve those cognitive demands while replacing regulated stainless-steel instruments with safe, cheap, tractable tokens.

8.5.2 Why AprilTag over QR Codes

QR codes are a natural alternative because they are widely available, readable by consumer devices, and familiar to most users. However, they are designed for data encoding rather than computer-vision pose estimation and marker tracking [96]. A QR code embeds dozens to hundreds of bytes of payload data, decodes it through a Reed-Solomon error-correction pipeline, and returns a string. AprilTag inverts that priority: it encodes a compact integer identifier using a binary matrix optimized for fast corner detection, rotation disambiguation, and error identification under perspective distortion, partial occlusion, and varying lighting [97]. Three practical differences make AprilTag the better choice for this prototype. First, detection speed and robustness. AprilTag exploits its known black border and grid structure to extract corners at sub-pixel precision. The OpenCV `aruco` module (which natively supports AprilTag dictionaries) can detect and decode dozens of markers in a single video frame in milliseconds, while QR-code detection requires finding three finder patterns, performing perspective correction, and decoding a payload — a heavier pipeline designed for different goals [98]; [99]. Experimental comparisons have shown that AprilTag achieves higher detection rates than QR codes under motion blur, shadow, and perspective distortion in pose-estimation contexts [100]. Second, compact dictionaries with guaranteed Hamming-distance bounds. AprilTag uses a lexicode-based coding system that guarantees a minimum Hamming distance between every pair of markers under all four rotations [97]; [101]. A high inter-marker distance means the detector can confidently identify a marker even when several bits are corrupted by glare, print quality, or partial occlusion. For a training prototype that must not confuse a learner’s correct selection with a detection error, that robustness is critical. Third, pose estimation. A single AprilTag marker provides four coplanar corner correspondences, sufficient for full 6-DOF pose estimation from a single calibrated camera [97]. Although the current prototype does not yet use pose for detailed spatial feedback, the same markers can later support richer interaction — detecting which instrument a learner hesitates over, flagging instruments placed in the wrong tray zone, or verifying the physical arrangement against a placement template. QR codes do not offer comparable corner-localization accuracy because their finder patterns are designed for barcode scanning rather than metric pose estimation.

Table 16: AprilTag versus QR code comparison

Requirement	QR code	AprilTag	Effect on TrayGuard
Detection under perspective, partial occlusion, and lighting variation	Moderate; designed for fronto-parallel scanning	Strong; designed for arbitrary camera pose and varying conditions	Fewer detection failures that inflate learner error rates [100].
Sub-pixel corner accuracy	Not designed for pose estimation	Native: lexicode-guaranteed dictionary with sub-pixel refinement	Enables future tray-zone and placement feedback [99].
Compact identifier with error correction	Large payload, Reed-Solomon on data bytes	Compact ID with lexicode Hamming-distance guarantee	Lower decode latency; no ambiguity between marker ID and payload parsing [97].
Open-source library in existing stack	OpenCV and platform libraries	Integrated in OpenCV <code>aruco</code> module; standalone C library	Zero additional dependency cost [96].
Neutral card front for assessment	QR pattern is visually distinctive and recognizable as a code	Binary grid is less obviously a machine-readable code to novices	Lower risk of learners recognizing markers as answer keys during assessment.

8.5.3 Why AprilTag over ArUco

ArUco is the closest alternative: both produce binary square fiducial markers, both are open source, and both can be detected through the same OpenCV `aruco` module. For this project’s requirements — up to 1000 markers and high detection accuracy — AprilTag is the stronger choice for three reasons. First, AprilTag’s lexicode system provides mathematical guarantees that ArUco’s search-based dictionary does not. The lexicode algorithm generates a family of codewords with a proven minimum Hamming distance under every possible rotation [97]; [101]. This means the detector can not only detect but also correct more bit errors per marker than ArUco’s heuristic dictionary of equivalent size. A larger minimum distance directly improves detection accuracy in the conditions most likely to occur during a training session: glare off laminated cards, low tabletop lighting, partial card occlusion, and off-angle camera views. Second, AprilTag’s corner-refinement pipeline (especially AprilTag 2’s adaptive thresholding and edge refinement) produces more stable corner localization across varying lighting and image scales [101]. The original AprilTag was already benchmarked as the most accurate fiducial system in head-to-head comparisons under motion blur and perspective distortion [100]. Third, AprilTag 3 introduces a multi-threaded detector and flexible tag layouts that improve throughput on higher-resolution images without sacrificing detection rate [102]. For the first build, the recommended family is `tagStandard52h13`, which provides 52 data bits and a minimum Hamming distance of 13 — enough error-correction capacity to support thousands of unique markers while still fitting comfortably on a printed card. If fewer than 500 markers are needed, `tagStandard41h12` is a more compact alternative with comparable robustness. Both families are supported through the AprilTag dictionary support in OpenCV’s `aruco` module, with full backward compatibility for earlier dictionaries (`DICT_APRILTAG_36h11`, `DICT_APRILTAG_16h5`), so no additional library dependency is required.

8.5.4 Why Not Real Instruments

The most obvious alternative — and the one that keeps the project closest to its original CV-checker goal — is to use real surgical instruments with a camera-based object detector. That path is rejected for the first prototype for four reasons. First, real instruments require regulated access, cleaning and handling

protocols, storage space, and educator supervision that the first study cannot assume [27]. Second, instrument recognition across manufacturers, wear states, and lighting conditions is an unsolved research problem that would dominate the engineering schedule and make the project claim fragile to a single detection failure [55]; [58]. Third, a real-instrument prototype would be difficult to reproduce across sites, since each hospital has different inventory, variant mix, and replacement cycles. Fourth, the cognitive task of interest — applying a local tray rule to select, count, and discriminate among items — can be studied without the sterility, weight, and handling requirements of real steel. Printed cards preserve the selection, count, and lookalike-discrimination demands while stripping away the confounding variables of instrument availability, cleaning policy, and physical handling risk. The card interface follows the same abstraction strategy as NeuroVase, which uses tangible cue cards to teach neurovascular anatomy and stroke assessment without requiring cadavers, imaging workstations, or clinical placements [80]. QR-code-based medical education tools similarly demonstrate that low-cost printable codes can support healthcare learning, simulation, and training access in resource-constrained settings [81]. The AprilTag card is a deliberate methodological choice that controls for instrument-access confounds while keeping the measured task aligned with the real cognitive demands of tray reconstruction.

8.6 Rationale for Gamified and Simulation-Based Learning Style

TrayGuard embeds retrieval practice, feedback, scoring, progressive difficulty, and confidence capture within a structured learning session that resembles a serious tabletop training game. This section justifies that design against traditional lecture-based training, video-only instruction, and unguided self-study.

8.6.1 Why Gamification and Simulation over Traditional Instruction

Traditional sterile-processing training relies heavily on lecture, video, and supervised on-the-job practice [29]; [27]. Those methods are necessary but not sufficient: lectures can introduce instrument names and tray rules, but they do not exercise the applied discrimination and count-sheet following that defines tray reconstruction. Simulation-based medical education with deliberate practice has been shown to outperform traditional clinical education in meta-analysis, and technology-enhanced simulation improves knowledge, skills, and behaviors compared with no intervention [71]; [72]. The features that make simulation effective — feedback, repetitive practice, curriculum integration, and measurable outcomes — map directly to TrayGuard’s learning loop [73]. Gamification adds a second layer. A 2021 systematic review of gamification in health professions education found that combining assessment attributes (scores, progress tracking) with conflict/challenge attributes (timed tasks, increasing difficulty) increased learners’ engagement with study material and led to improved learning outcomes in most studies that included a comparison condition [103]. A more recent umbrella review of game-based learning in nursing education found a large pooled effect on academic knowledge performance (standardized mean difference 1.06) and a moderate effect on skill performance (0.54) [104]. The mechanism is plausible: game elements increase repetition, focus attention on error patterns, and make low-stakes practice more acceptable than high-stakes testing. The same review cautions that most gamified intervention studies lack well-defined control groups and theoretical grounding for their choice of game elements [103]. TrayGuard addresses that concern by grounding each game-like feature in a specific learning mechanism.

8.6.2 Specific Game Elements and Their Learning Rationale

Table 17: Game elements and their learning-science rationale

Game-like feature	Learning mechanism	Evidence base
Pre-test establishes a baseline score	Goal-setting and self-assessment calibration before practice	Pre-testing improves later retention compared with studying without a prior test, even when answers are not provided [74]; [75].
Retrieval cards use prompt-before-reveal	Active recall with immediate confirmation; the core retrieval-practice effect	Tests improve long-term retention more than restudy across health-professions contexts [76]; [78].
Quiz with score and weak-item logging	Retrieval with spaced repetition and targeted remediation	Weak-item repetition reduces error persistence on later tasks [28]; [79].
Practice tray sort with error-category feedback	Deliberate practice with specific, actionable correction	Immediate error-specific feedback improves skill acquisition more than correctness-only feedback [79]; [73].
Confidence capture after each item	Metacognitive calibration; exposes high-confidence errors	High-confidence errors are more resistant to correction and indicate overconfidence that feedback must target explicitly [28].
Parallel no-hints post-test	Near-transfer assessment without answer leakage	Separating assessment from practice prevents score inflation from memorization of exact layouts [105].
Metrics export and weak-item summary	Learner- and instructor-facing performance data for targeted remediation	Instructors cannot target weak items or weak rules without item-level evidence [30]; [3].

Each element serves a specific cognitive purpose rather than decoration. Points and progress bars (the most common gamification elements in medical education) are present only insofar as they support self-assessment: the pre/post accuracy score, the weak-item recovery count, and the error-category breakdown are the central feedback signals. The design deliberately avoids leaderboards, badges, and competitive ranking, which can shift motivation from intrinsic learning to extrinsic reward-seeking in health-professions contexts [103].

8.6.3 Why Not a More Complex Game Format

Several alternative gamification approaches were considered and rejected. Full virtual reality (VR) simulation offers immersive instrument handling but requires hardware (headset, controllers), room space, onboarding time, and technical support that the first prototype cannot assume. SteriBoost, a commercial VR sterile-processing trainer, demonstrates the feasibility of VR for this domain, but its hardware and custom-scene requirements place it at a later validation stage rather than the first project prototype [106]. Competitive multiplayer formats (leaderboards, team challenges) can increase engagement but risk publicizing individual error rates in a training context, potentially creating the punitive environment that the ethics controls explicitly prohibit (Section 15). They also require larger participant pools and session coordination that the first 8-12-person study cannot support. Simulated patient scenarios or narrative-driven learning modules (used in tools such as Touch Surgery) can improve engagement and contextual reasoning, but they add authoring burden and reduce the number of tray-repetition cycles per session [88]; [89]. TrayGuard’s target task — learning a specific local tray rule — benefits more from

repeated applied practice than from narrative context, so the game structure stays close to the assessment loop. The chosen gamification level is therefore lightweight, mechanism-driven, and designed around the retrieval-practice and feedback evidence. It is closer to the serious-game approach of SteriDefi, the SF2S sterilization serious game that uses question banks, difficulty levels, and scoring to motivate repeated knowledge retrieval, than to a narrative or VR simulation [107].

8.6.4 Learning Mechanisms Behind the Game Elements

Structured, repeated practice with immediate, error-specific feedback, delivered in a bounded session with clear goals and measurable progress, produces measurable short-term learning. Game elements sustain the repetition that retrieval-practice research shows is necessary for durable learning [74]; [76]. The "game" is the learning loop itself — the same logic that makes flashcard apps, spaced-repetition quizzes, and simulation-based skills trainers effective teaching tools. A learner who improves on the TrayGuard simulated tray task has undergone repeated retrieval practice on local names, aliases, counts, and lookalike distinctions, followed by an applied sorting task with category-specific feedback.

9 Requirements Definition

9.1 Element Definition

Table 18: Element definition for the TrayGuard system

Element	Definition For TrayGuard
Intended practice	Novice preparation for local tray reconstruction before or alongside supervised SPD training.
Other practice affected	Instructor review, preceptor coaching, tray content updates, weak-item remediation, and future CV-supported tray review.
Artifact	A local training app, file-backed tray module, printable marker cards, tray-sorting surface, scoring engine, and export/report package.
Problem addressed	Learners need repeated, measurable practice applying local tray rules, counts, aliases, and lookalike discrimination before real tray work.
Technology	Local web/app workflow, JSON tray modules, QR or AprilTag cards, camera or manual selection input, scoring logic, and CSV/JSON export.
Primary users	Novice learners, instructors, SPD educators, and future project teams.
Perception goal	The system should feel like a serious practice tool, not a clinical approval device or punitive surveillance system.
Environment	Classroom, skills lab, project demo, or supervised training setting; not a live sterile field or production SPD deployment.
Core function	Convert a local tray module into study, quiz, simulated sorting, feedback, assessment, and learner metrics.
Required behavior	Separate practice from assessment, suppress hints during pre/post modes, classify errors consistently, and export comparable rows.
Structure	Content module, learning workflow, marker/manual input, scoring engine, feedback engine, reporting/export, and instructor review.
Intended effects	Improve simulated tray-sorting accuracy, reduce severe errors, expose weak items, calibrate confidence, and reduce early preceptor burden.
Possible side effects	Overclaiming simulated scores, punitive use of learner metrics, answer leakage, inequitable access to updated modules, and reduced attention to supervised hands-on practice.

9.2 Objective Tree

Improve novice readiness for local tray reconstruction through repeatable, measurable simulated practice.

1. Increase local tray knowledge
 - 1.1 Teach local names, aliases, and instrument families

- 1.2 Teach required counts and tray-specific rules
- 1.3 Teach lookalike and same-family distinctions
- 2. Improve applied tray-sorting performance
 - 2.1 Reduce missing required items
 - 2.2 Reduce wrong and misidentified items
 - 2.3 Reduce extra and wrong-count items
 - 2.4 Maintain reasonable task time without rewarding guessing
- 3. Make learning measurable
 - 3.1 Capture pre-test and post-test accuracy
 - 3.2 Export item-level error categories
 - 3.3 Capture confidence, Not sure choices, and high-confidence errors
 - 3.4 Preserve comparable assessment variants
- 4. Support instructors
 - 4.1 Show weak instruments and weak tray rules
 - 4.2 Keep modules local, versioned, and reviewable
 - 4.3 Avoid collecting patient data or unnecessary learner identifiers
- 5. Preserve claim boundaries
 - 5.1 Label outputs as simulated practice evidence
 - 5.2 Require expert review before content-valid claims
 - 5.3 Require SPD pilot evidence before workplace-effectiveness claims

9.3 Performance Specifications

The first build should use the following targets. They are deliberately formative rather than clinical: they define when the prototype is ready to support a project novice-learning claim.

Table 19: Performance specifications and acceptance targets

Area	Target	Evidence Basis	Measurement
Participant count	Minimum 8 novice participants; target 12 complete participants if schedule allows.	NeuroVase used a 40-participant controlled study for an educational card/AR system, while qualitative methods literature supports small, narrow, homogeneous samples for early formative work [80]; [108]; [109].	Unique participants with complete pre/post attempts.
Completion rate	At least 7 of 8 participants, or 85% when $n > 8$, complete the full loop without blocking moderator intervention.	Usability must be separated from learning; NeuroVase used SUS and user-experience measures alongside knowledge testing.	Completion flag and blocking help requests.
Accuracy gain	Mean paired tray accuracy improves by at least 20 percentage points from pre-test to post-test.	Ofstead's SP training pilot showed large pre/post score movement after structured teaching and hands-on practice; TrayGuard should set a smaller but visible formative threshold [28].	<code>post_accuracy - pre_accuracy</code> .
Post-test floor	Mean post-test accuracy is at least 80%, and no more than one participant scores below 70%.	The goal is not only improvement from a low baseline; learners should reach a usable simulated familiarity level before the module is considered ready.	Post-test <code>accuracy_score</code> .
Severe-error reduction	Mean <code>missing + wrong + misidentified</code> errors decrease by at least 30%.	Alfred et al. and Nichol et al. show missing, wrong, extra, and assembly-related errors are operationally meaningful categories [1]; [3].	Paired error-category counts.
Speed without unsafe guessing	Median post-test duration is at least 10% lower, or is no more than 20% higher if accuracy improves by at least 20 points.	Time matters because tray assembly is workflow-bound, but speed alone could reward guessing.	<code>duration_seconds</code> , paired by learner.
Confidence calibration	High-confidence errors decline by at least 30%; low-confidence correct answers are exported.	Ofstead captured confidence/satisfaction, and feedback research supports showing learners where they are and what to do next [28]; [79].	Confidence ≥ 4 on incorrect items; confidence ≤ 2 on correct items.
Retention	A 7-day delayed tray sort retains at least half of the immediate accuracy gain and delayed accuracy does not return to within 5 percentage points of the pre-test baseline.	Bell et al. showed immediate medical-learning gains can decay within days and recommended reinforcement after as little as 1 week; same day	Delayed <code>accuracy_score</code> relative to pre/post.

9.4 Quality Function Deployment

Compact QFD:

Table 20: Quality function deployment: needs to engineering targets

Stakeholder need	Engineering characteristic	Target value	Verification
Learners can practice local tray content repeatedly.	File-backed module with required items, distractors, aliases, features, counts, and variants.	One complete basic_general_tray_v1 module with 8-12 required concepts and 3-6 distractors.	Module schema review and demo load.
Practice feels connected to tray reconstruction, not isolated memorization.	Simulated tray sort using physical cards or manual selection.	Pre-test, practice, post-test, and optional delayed variant all use the same tray rule.	End-to-end run.
Educators can see what learners missed.	Attempt-level and item-level export.	100% of completed runs export all required fields.	Schema validation.
Errors map to operationally meaningful categories.	Scoring engine with missing , extra , wrong , misidentified , and wrong_count .	Every submitted tray receives category counts.	Unit tests or scored example cases.
Assessment is not contaminated by coaching.	Mode boundaries and feedback suppression.	No hints or corrective feedback in pre-test, post-test, or delayed modes.	UI walkthrough and export mode check.
The physical marker layer does not corrupt scores.	Marker reliability and user confirmation.	At least 95% detection in normal layouts; confirmation before scoring.	Study 0 marker log.
Learner confidence is visible.	Confidence or Not sure capture.	High-confidence errors and low-confidence correct answers exported.	Export inspection.
The system remains ethical and nonpunitive.	Privacy and report labels.	Hashed learner IDs, no patient data, and report language that says practice evidence.	Report review.
Future teams can extend the design.	Versioned data model and traceability.	Module version, source notes, verification status, and assessment variant IDs included.	Data file review.

9.5 Requirement Traceability Matrix

Table 21: Requirement traceability matrix

Req.	Stakeholder need	Design feature	Evidence source	Validation method	Status
FR-1 Load local module	Educators need updateable local tray rules.	JSON tray module and module selector.	Count-sheet and local tray evidence [11].	Change module data and reload.	Ready for impl.
FR-2 Retrieval cards	Learners need active recall of names, features, and counts.	Prompt-before-reveal cards.	Retrieval and flashcard evidence [74]; [78].	Card walkthrough and quiz logs.	Ready for impl.
FR-3 Pre/post assessment	Reviewers need measurable learning evidence.	No-hints pre-test and post-test variants.	Ofstead pre/post pattern [28].	Paired run export.	Protocol-ready.
FR-4 Quiz mode	Learners need feedback before applied sorting.	Quiz with confidence and weak-item logging.	Feedback and retrieval evidence [79].	Quiz result export.	Ready for impl.
FR-5 Practice feedback	Learners need actionable correction.	Error-specific feedback engine.	Tray defect categories [1]; [3].	Seeded error cases.	Defined.
FR-6 Metrics export	Instructors need reviewable evidence.	Attempt and item CSV/JSON.	Error-reporting and competency evidence [3]; [28].	Schema validation.	Defined.
FR-7 Mode separation	Assessment must remain fair.	Mode state and feedback suppression.	Assessment design and answer-leakage risk.	UI walkthrough.	Defined.
NFR-1 Fast data load	Demo cannot stall.	Local file storage.	Usability expectations.	Load timing ≤ 2 seconds.	Future test.
NFR-2 Short flow	Participants can finish in one session.	Bounded module and timed segments.	Study protocol and sample feasibility.	Timing ≤ 20 min for demo.	Future test.
NFR-3 Export reliability	No completed run should be lost.	Local export after each run.	Study evidence requirement.	100% completed-run exports.	Future test.
NFR-4 Privacy	Learners should not be surveilled unnecessarily.	Hashed IDs and no PHI.	Healthcare data governance concerns.	Export review.	Defined.
NFR-5 Content traceability	Local content must be auditable.	Source notes and instructor verification status.	Count-sheet and local content boundary.	Module review.	Defined.
NFR-6 Capability labels	Stakeholders must not overread scores.	Report language: simulated practice evidence.	Certification and clinical-transfer boundaries.	Report review.	Defined.

10 System Design

10.1 System Overview

TrayGuard should be built as a local module runner plus a card-scanning assessment surface. The app does not need to recognize real instruments for the first study. It needs to observe which simulated instrument cards the learner placed in the tray area, score that selection against a local tray template, and export evidence.

```
local tray module
-> pre-test AprilTag tray sort
-> retrieval-first study cards
-> quiz with confidence / Not sure
-> practice AprilTag tray sort with feedback
-> post-test AprilTag tray sort
-> metrics export
-> committed 7-day delayed retention sort (see falsification criteria in Section~\ref{sec:study-2})
```

The architecture follows the evidence base:

- Local tray modules are required because count sheets define tray names, contents, quantities, sizes, and reference/catalog numbers, and real count sheets include fields such as reference number, description, location, quantity, check box, total count, and hospital signature [11]; [69]; [70].
- Retrieval-first study and quiz modes are justified by retrieval practice, health-professions flashcard evidence, and spaced digital education [74]; [78]; [77].
- Pre/post practice, confidence capture, and optional delayed retention follow the sterile-processing precedent in Ofstead and the retention warning in Bell et al. [28]; [105].
- AprilTag cards are a physical proxy, not a real-instrument detector. The tangible-card idea is supported by NeuroVase-style cue cards and board/card-based healthcare education, but clinical transfer remains future validation [80]; [84].

10.2 Subsystem Breakdown

The system separates into these functional subsystems:

Table 22: Functional subsystem breakdown

Subsystem	Responsibility
Local content/tray module	Stores instrument records, aliases, photos, distinguishing features, required quantities, distractors, and tray-template versions.
Learning workflow	Routes the learner through pre-test, study, quiz, practice sort, weak-item review, post-test, and export.
Retrieval practice	Presents prompt-before-answer cards and quizzes for instrument names, features, lookalikes, and required counts.
AprilTag card scanning	Detects printed marker cards and maps marker IDs to instrument IDs so the app can observe a physical tray simulation.
Simulated tray scoring	Compares detected cards and selected quantities against the tray template.
Feedback engine	Reports missing, extra, wrong, misidentified, and wrong-count errors in practice mode.
Confidence/uncertainty capture	Records confidence or Not sure responses for assessment and instructor review.
Reporting/export	Produces pre/post accuracy, time, error-category, confidence, and weak-item summaries.
Instructor review	Lets an educator verify tray content, local naming, distractors, and study evidence before broader use.

Computer vision for recognizing real instruments remains a future support subsystem. In the project build, the marker-scanning subsystem is deliberately a proxy for observing user decisions, not a claim that the app can identify real surgical instruments under clinical conditions.

10.3 Data Model and Module Contract

The data model should mirror a real count sheet while staying small enough for the first build. Stryker’s public count sheets show the practical minimum: reference number, description, location/layer, quantity, check box, total count, additional items, and hospital signature. TrayGuard adds learning-specific fields for aliases, lookalikes, confidence, and error categories.

Table 23: Data model objects and required fields

Object	Required Fields	Why It Matters
Instrument	id, display_name, family, aliases, distinguishing_features, image_refs, apritag_id, local_verification_status	Supports retrieval cards, quiz prompts, card scanning, and local naming.
TrayTemplate	id, name, version, procedure_family, source_note, required_items, distractor_items, lookalike_pairs, assessment_variants	Stores the local count-sheet task and separates required instruments from distractors.
TrayItemRequirement	instrument_id, quantity, reference_number, location_or_layer, acceptable_substitutes, notes	Mirrors real count-sheet fields and makes count/location errors measurable.
AssessmentVariant	id, mode, random_seed, required_item_ids, distractor_item_ids, card_front_policy, feedback_enabled	Keeps pre/post/delayed tasks comparable while preventing answer leakage.
LearningRun	run_id, learner_id_hash, participant_group, tray_template_id, variant_id, mode, started_at, completed_at	Separates study, quiz, practice, pre-test, post-test, and delayed attempts.
DetectedCard	run_id, apritag_id, instrument_id, timestamp, detection_confidence, confirmed_by_user	Lets marker failures be audited instead of silently counted as learner errors.
AttemptResult	run_id, selected_items, duration_seconds, accuracy_score, required_recall, errors, confidence_summary	Provides the main pre/post evidence.
ItemResult	run_id, instrument_id, required_qty, selected_qty, error_category, confidence, not_sure	Supports item-level feedback, weak-item review, high-confidence error reporting, and instructor review.

The first implementation can store these objects in JSON files and export CSV for analysis. The important constraint is schema stability: pre-test, post-test, and delayed retention must produce comparable rows. Sample instrument JSON:

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo scissors, 5.5 in",
  "family": "cutting_dissecting",
  "aliases": [
    "straight Mayo",
    "suture scissors"
  ],
  "distinguishing_features": [
    "Heavy straight blades",
```

```

    "Broader blade profile than Metzenbaum scissors",
    "Ring-handled cutting instrument"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55",
    "metzenbaum_scissors_curved_55"
  ],
  "image_refs": [
    {
      "id": "view_a",
      "path": "data/instruments/mayo_scissors_straight_55_view_a.jpg",
      "source": "local_photo",
      "approved_for_study": true,
      "approved_for_assessment": true,
      "attribution": "Local TrayGuard photo"
    },
    {
      "id": "card_front_neutral",
      "path": "outputs/cards/mayo_scissors_straight_55_neutral_front.png",
      "source": "generated_card",
      "approved_for_study": false,
      "approved_for_assessment": true,
      "attribution": "TrayGuard AprilTag card"
    }
  ],
  "apriltag_id": 12,
  "card_front_policy": "neutral_assessment_front",
  "study_prompts": [
    {
      "id": "name_recall",
      "prompt": "Name this instrument.",
      "answer": "Straight Mayo scissors, 5.5 in"
    },
    {
      "id": "feature_recall",
      "prompt": "What feature helps distinguish this from Metzenbaum scissors?",
      "answer": "Straight Mayo scissors have heavier, broader blades."
    }
  ],
  "notes": "Use local photos before pilot. Do not rely on vendor images for assessed study content.",
  "local_verification_status": "pending_review"
}

```

10.4 Mode Behavior

Table 24: Training mode behavior by hints, feedback, and metrics

Mode	Hints	Corrective Feedback	Metrics	Use
Pre-test	No	No	Yes	Baseline simulated tray-task proficiency.
Study cards	Yes	Yes	Optional	Prompt-before-reveal retrieval of names, aliases, families, features, and counts.
Quiz	Limited after answer	Yes	Yes	Retrieval practice, confidence calibration, and weak-item discovery.
Practice sort	Yes	Yes	Yes	Applied retrieval and deliberate practice on tray organization.
Post-test	No	No	Yes	Comparable same-day assessment after practice.
Delayed retention	No	No	Yes	Checks whether gains persist after 7 days (with falsification criteria).

10.5 Learning Loop

The product is one learning loop:

```
local tray module -> pre-test -> retrieval-first cards -> quiz -> practice tray sort -> post-test ->
  metrics export
```

Retrieval practice is the instructional engine. The learner should usually commit to an answer before the system reveals it: name the instrument, identify a distinguishing feature, choose the lookalike, recall the count, or decide whether the item belongs in the tray. The simulated tray sort is applied retrieval: the learner must use those same facts in context with distractors and quantities.

Each feature in the learning loop maps to a specific evidence-based role:

Table 25: Feature-to-evidence mapping for the learning loop

Feature	Evidence-based role	Engineering requirement
Retrieval-first study cards	Initial instruction that still asks the learner to answer before reveal	Store photos, names, aliases, functions, distinguishing features, and local notes; present prompts before explanations
Identification quiz	Central retrieval practice for instrument identity and features	Ask recall or recognition questions, randomize distractors, record correctness, confidence, and time
Simulated tray sorting	Safe applied retrieval in a realistic work task	Let learners assemble or check a tray against a count sheet in a low-risk training environment
Immediate error-specific feedback	Deliberate practice and error correction	Explain wrong answers and classify errors as missing, extra, wrong, misidentified, or wrong count
Confidence ratings	Metacognitive signal for learner and instructor	Capture confidence before or after tasks so low-confidence correct answers and high-confidence errors are visible
Separated pre/post tests	Evaluation not contaminated by coaching	Disable hints and feedback during tests; export comparable pre/post accuracy, time, error type, and confidence

10.6 Module Scope

The first module should contain:

- 8-12 required instrument concepts;
- 3-6 distractor instruments;
- 2-4 known lookalike or same-family pairs;
- local names, aliases, family labels, required counts, and distinguishing features;
- AprilTag cards that represent instruments during the physical tray simulation;
- parallel pre-test, post-test, and delayed-retention tray variants with matched difficulty;
- instructor or expert review before results are presented as content-valid.

The cards should use neutral assessment fronts. Names, counts, and obvious answer labels should be hidden during pre-test, post-test, and retention tasks so the participant cannot solve the task by reading the card.

10.7 FGVC-Derived Module Plausibility

The FGVC-derived modules are practice modules built from the major-tray image classes reported by Atabuzzaman et al. [56]. The source contributes instrument names and images for a bounded image-card set: scissors, clamps, forceps, needle holders, retractors, suction, and other classes present in the major-tray archive. It gives TrayGuard a coherent pool of real instrument classes photographed under one dataset program.

Plausibility for these modules comes from three linked sources of evidence. First, every required card, distractor card, and lookalike pair is selected from the FGVC major-tray class mapping. Second, the module schema mirrors real tray/count-sheet fields: tray name, required item, quantity, reference number, location or layer, total count, and check status [69]; [70]. Third, the module size follows the project study constraint: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs, so novices can complete pre-test, practice, post-test, and delayed retention tasks in a single session. The FGVC-derived quantities are simulated as one card per selected class. Expert review sets the final clinical wording, counts, substitutes, and local tray membership before an SPD pilot uses the module as content-valid tray material.

10.8 Real Tray List Examples

The first TrayGuard module should be smaller than a clinical tray, but its data fields should look like real tray/count-sheet data. Three examples are useful:

Table 26: Real tray list examples and design implications

Real Example	What It Shows	Design Implication
Stryker Gamma4 Indication Tray count sheet	A real vendor count sheet uses tray name, insert/base sections, reference numbers, descriptions, locations, quantities, check boxes, total instrument counts, additional items, and hospital signature [69].	TrayGuard should store reference_number , description/display_name , location_or_layer , quantity , and additional_items fields even if the first module only uses names and quantities.
Stryker IMN Basic Instruments Tray count sheet	A second real count sheet uses the same pattern and splits the tray into insert/base groupings with total counts of 15 and 10 instruments [70].	The data model should support sections/layers and total-count validation, not just a flat list.
Published major orthopedic tray study	Toor et al. observed an 88-instrument major orthopedic tray across 80 procedures and compared optimized tray configurations of 47, 67, and 51 instruments [112].	Real clinical trays can be far larger than the first module. The first study should use 8-12 required concepts for feasibility, while preserving fields that can scale to larger trays.

For the first build, use a teaching-scale tray: 8-12 required concepts, 3-6 distractors, and 2-4 lookalike pairs. This is intentionally smaller than the real examples so a novice can complete the loop in 45-60 minutes.

11 Analysis

This section consolidates the analysis that demonstrates the capabilities of the final training-centered design.

11.1 Prototype Capability Demonstration

The following capabilities have been demonstrated in the built prototype:

Table 27: Demonstrated prototype capabilities

Capability	Evidence	Status
Tray module loading	<code>basic_general_tray_v1</code> module loads and renders in the web app	Verified
Retrieval-first study cards	Prompt-before-reveal cards display instrument names, images, aliases, features, and tray quantities	Verified
Identification quiz	Quiz presents instrument images and captures name, confidence, and Not sure responses	Verified
AprilTag card detection	Camera-based detection maps marker IDs to instrument IDs in real time; manual entry fallback available	Verified
Simulated tray scoring	Submitted selections scored against tray template with accuracy and error categories	Verified
Error-category classification	Missing, extra, wrong, misidentified, and wrong-count errors computed per run	Verified
Pre/post assessment mode	No-hints mode with matched variants; baseline and post-training evidence collected	Verified
Confidence capture	1–5 scale and Not sure checkbox recorded per item; high-confidence errors flagged	Verified
CSV/JSON export	Attempt-level and item-level records exported with all required fields	Verified
Card generation	CLI generates printable marker PNGs, grid index sheets, and double-sided study flashcards	Verified

Every verified capability supports a direct link to the performance specifications (Section 4.3) and the requirements traceability matrix (Section 9).

11.2 Marker Reliability Analysis

The marker layer is a measurement tool: if detection fails, scores confound learner error with app error. The following analysis frames the reliability expectation:

Table 28: Marker reliability analysis

Dimension	Target	Method
Detection rate in normal layouts	$\geq 95\%$	Bench test with varied card spacing, rotation, and lighting
Detection rate in degraded conditions	Reported as limitation	Glare, partial occlusion, and extreme angles logged per test
Marker-ID confusion rate	0% in controlled test	All markers in the printed set decoded against ground truth
False positive rate (non-marker objects)	Reported as limitation	Objects placed in camera view that do not contain markers
User confirmation before scoring	Required	Confirmation screen shows detected cards for user review before scoring

The target values match the performance specification (Section 4.3) and the QFD requirement for marker reliability. Study 0 in the evaluation plan (Section 13.5) provides the experimental protocol for formal measurement.

11.3 Unit Test Results

Unit tests verify core logic that directly affects learner evidence quality:

- **Module loading.** JSON schema validation confirms that all required fields are present and correctly typed for the primary tray module.
- **Scoring accuracy.** Seeded tray submissions with known error patterns (one missing, one extra, one lookalike substitution, one wrong-count) return the expected category counts and overall accuracy score.
- **Export completeness.** CSV and JSON exports from scored runs contain all required fields at both the attempt level and item level.
- **Card generation.** Generated AprilTag PNGs decode to the correct marker IDs, and the double-sided flashcard layout renders instrument data without text overflow.

11.4 Design Traceability

Every design decision in the training path traces to an objective in the objective tree (Section 4.1), a specification in the performance specification analysis (Section 4.3), and a requirement in the requirement traceability matrix (Section 9). The research support audit (Appendix A) provides choice-by-choice traceability to published literature, and the implementation defaults (Appendix B) document the critique-ready interpretation of each current implementation decision.

12 Design Outcome versus Requirements – Comparison

This section evaluates the original computer-vision tray checker design against the requirements it was intended to satisfy. The CV path is treated as a documented engineering effort that produced useful technical evidence but did not achieve deployment-grade reliability, leading to the training-centered pivot documented in Section 8.

12.1 CV Requirements Assessment

The original tray-verification concept defined requirements for automated tray-status logic, technician-facing UI, physical capture fixture, and detection reliability. Each requirement is assessed below.

Table 29: CV design outcome versus requirements

Requirement	Target	Achievement	Gap
Tray-status logic with two-threshold confidence band	Present, Needs Review, Missing, Rescan states	Logic specified and documented	Not tested with real SPD workflow data
Technician-facing UI with one primary action per state	Clear hierarchy, fast correction, specific review reasons, human sign-off	UI requirements specified	Not implemented beyond design documentation
Physical capture fixture	Overhead camera on portable weighted board	Fixture design documented	Not fabricated or tested in SPD conditions
Lighting robustness	Detection maintains mAP50 above 0.7 across SPD-like illumination range	Evaluated across 800–80 lumens; overlay layout caused largest drop	mAP50 below 0.3 in overlay conditions; lighting still a deployment barrier
Lookalike discrimination	Correctly distinguish straight/curved, narrow/broad same-family pairs	Evaluated with 6 instrument-like classes	Pairwise confusion above 30%; high-confidence wrong predictions persisted
Open-set rejection	Detect unknown objects as unfamiliar rather than assigning spurious labels	Evaluated with unknown-similar distractors	Unknown false positive rate 40–60%; model did not reject unfamiliar objects
Synthetic-to-real transfer	Synthetic pretraining enables few-shot real deployment	5–10 pp mAP50 improvement over scratch	Absolute mAP50 on real test set below 0.70

12.2 Gap Analysis and Pivot Justification

The cumulative evidence showed that CV detection was reliable only in matched, controlled conditions and degraded under every variation that SPD workflows routinely present: variable lighting, reflective backgrounds, overlapping instruments, lookalike variants, and unknown items. Three structural gaps prevented the project from closing to a deployment-grade system:

1. **Data coverage.** Real SPD deployment would require site-specific training data covering local instruments, lighting conditions, tray layouts, and manufacturer variants. The project had access to only a small proxy dataset.
2. **Failure-mode validation.** A clinical tool needs validated behavior for glare, occlusion, blur, unknown objects, class confusion, false alerts, and human review integration. Each mode requires its own experiment, acceptance criterion, and risk assessment.
3. **Workflow integration.** The tray checker must fit into SPD production flow, device policy, cleaning protocols, and privacy requirements – all outside the project’s available clinical access.

The training-centered pivot preserved the project’s technical contributions (the CV experiments, data collection pipeline, and detection code are documented in Section 6) while replacing the main evidence path with a claim that was buildable, testable, and interpretable within the project’s resources. The CV infrastructure remains available as a future support layer after separate validation (Section 6.5).

13 Evaluation And Validation Plan

13.1 Evaluation Framework

The question of how to prove that people have learned requires an evaluation framework that makes explicit what counts as evidence, for which claim, and at what level of certainty. The training-evaluation and educational-assessment literatures provide well-established structures for this.

Kirkpatrick’s four-level model is the most widely used framework for training evaluation: Level 1 (Reaction), Level 2 (Learning), Level 3 (Behavior), and Level 4 (Results) [113]. In this model, a pre/post accuracy gain on a simulated tray task is Level 2 evidence: it shows that knowledge or skill was acquired during the training session. The professor’s question about whether accuracy scores translate to real work corresponds to Level 3 (Behavior transfer), which requires measuring performance in or near the actual work context. Arthur et al.’s meta-analysis of organizational training confirms this structure: they found that the mean effect size for learning outcomes (Level 2) was larger than for behavior (Level 3) or results (Level 4), and that evaluation designs must match the claim level [82]. Salas et al. further argue that effective training should be treated as a system that analyzes work, targets specific competencies, provides practice and feedback, and evaluates transfer [83]. The current study ladder addresses Level 2 directly and builds toward Level 3 through convergent evidence and expert review, but does not claim Level 4 (reduced OR delays or hospital defect rates), which requires an SPD-site study.

Within educational assessment, Messick’s unified validity theory defines construct validity as a single concept supported by six aspects: content, substantive, structural, generalizability, external, and consequential [110]. The Standards for Educational and Psychological Testing adopt this unified view, treating validity as the degree to which evidence and theory support the intended interpretations of test scores [114]. Cook and Beckman operationalize the same framework for health-professions education as five sources of validity evidence: content, response process, internal structure, relations to other variables, and consequences [111]. Kane’s argument-based approach adds that validity depends on the plausibility of an interpretive argument that connects test scores to their intended use, with assumptions that must be explicitly warranted at each inference step: scoring, generalization, extrapolation, and implications [115].

Applied to TrayGuard, these frameworks require the following evidence chain:

1. **Content validity:** the tray module, names, aliases, counts, distractors, and lookalike pairs are plausible for a training tray. Addressed by expert review (Study 3) and instructor-verified module content.
2. **Response process:** learners interact with the system as intended; confidence ratings and **Not sure** capture reflect genuine self-assessment rather than confusion or guessing. Addressed by usability observation and confidence-calibration analysis (Studies 1-2).
3. **Internal structure:** error categories behave consistently: lookalike pairs produce misidentified errors, distractors produce extra errors, and repeated practice reduces specific error types. Addressed by pre/post error-type analysis (Studies 1-2).
4. **Relations to other variables:** TrayGuard tray-sort scores correlate with an independent measure of the same knowledge. This is the weakest link in the current design. Addressed by a convergent photo-identification test added to Study 1.
5. **Consequences:** the scores support the intended interpretation (improved simulated tray familiarity) and do not support unintended interpretations (clinical competence, SPD readiness). Addressed by scope boundaries and report labeling (Section 14).

A sixth issue spans the last two sources: **population generalizability**. Study 1 tests complete novices (college students with no SPD experience), whose pre-test accuracy is near zero. The target population

is SPD workers, whose baseline accuracy on familiar trays may already be high. A pre/post gain from 0 to 90 in novices does not, by itself, predict a gain from 90 to 90 in experienced workers. This is a classical range-restriction and ceiling-effect problem in training evaluation [82]. Kane’s extrapolation inference requires evidence that the learning mechanism transfers across populations even when the score distribution shifts [115].

There are three responses to this objection within the paper’s scope:

1. **The Ofstead precedent shows that experienced workers are not at ceiling on novel or specialized tasks.** Ofstead et al.’s sterile-processing training pilot used certified SP professionals whose mean pre-test score was 41% — well below ceiling — on a borescope visual-inspection task that was new to them [28]. SPD workers are experts on their current trays but will face new modules, updated count sheets, loaner trays, or specialized instrument sets. TrayGuard’s measurable value for an experienced population is not absolute accuracy gain (which may be small) but error-type reduction (eliminating the specific errors they do make), confidence calibration (reducing high-confidence errors), standardization across shifts (aligning names and rules), and retention maintenance (preventing skill decay between exposure to low-frequency trays).
2. **The dependent variable changes with the population.** For novices, the primary outcome is pre/post accuracy gain (baseline → learning). For experienced workers, the primary outcomes shift to: error-type reduction (fewer missing, wrong, misidentified items on known trays), speed improvement without accuracy loss, confidence calibration (high-confidence error rate), new-module learning rate (time to criterion on an unfamiliar tray), and retention over time (preventing decay on low-frequency trays). Study 4 (SPD pilot) must specify which dependent variable applies, rather than reusing the novice-study design unchanged [113].
3. **The learning mechanism is population-independent even if the effect size is not.** Retrieval practice, immediate error-specific feedback, and spaced repetition improve performance across learner levels in health-professions education [74, 76, 75]. The novice study establishes that the mechanism works in this task domain. The SPD pilot establishes the effect size for the target population. Arthur et al.’s meta-analysis confirms that training effect sizes vary by trainee population, which is why the study ladder separates the two populations rather than extrapolating effect sizes [82].

The study ladder (Table 30) maps each study level to the evidence source it earns. No single study provides all sources; the claim is supported by the accumulated pattern.

13.2 Current Evidence Available

The current evidence package includes:

- literature-backed problem definition for SPD tray reconstruction and assembly errors;
- support for simulation, retrieval practice, flashcards, feedback, tangible card interfaces, and board/card-based healthcare education;
- a bounded final product concept based on local tray modules and AprilTag instrument cards, with the evaluation framework above clarifying which claims each evidence source supports;
- a subsystem architecture, data contract, mode behavior, and scoring rubric;
- a user-study plan that addresses content validity, response process, internal structure, relations to other variables through convergent evidence, and retention through a committed delayed check.

13.3 Formative Novice Study Plan

The first feasible study is a formative novice study with students or other participants who do not have SPD tray experience. The study should evaluate whether the AprilTag-card simulation teaches a bounded local tray module, not whether participants are ready for real SPD work. Use one tray module with 8-12 required instruments, a distractor pool, and parallel pre/post tray variants. A reasonable first sample is 8-12 participants for classroom evidence; a larger follow-up can use paired statistics once the prototype and protocol are stable. Study sequence:

1. Consent and prior-experience questionnaire.
2. No-hints pre-test: assemble the simulated tray from AprilTag instrument cards using a tray list or count-sheet-style prompt.
3. Retrieval-first study cards for instrument names, aliases, distinguishing features, and counts.
4. Short quiz with confidence or **Not sure** capture.
5. Practice tray sort with immediate error-specific feedback.
6. Weak-item review for missed or uncertain instruments if time permits.
7. No-hints post-test on a comparable tray variant.
8. Short interview about confusing instruments, feedback usefulness, and perceived realism.

Primary measures are pre/post tray accuracy, duration, missing/wrong/extra/ misidentified/wrong-count errors, confidence calibration, high-confidence errors, low-confidence correct answers, and weak-item recovery. The study supports the training claim if most participants improve accuracy without a severe-error increase, maintain or improve completion time, and can explain at least one corrected error after practice.

The primary value of the novice study is **instrument validation**, not population-effect estimation. The study answers five prerequisite questions that apply to any subsequent population:

1. Do the error categories behave as expected (lookalike confusion vs. missing vs. extra items), or does the scoring rubric produce noise?
2. Does the convergent photo-identification test show a moderate-to-strong correlation with tray-sort accuracy ($r \geq 0.50$)? If not, the tray scores do not measure instrument knowledge in *any* population.
3. Is confidence calibration measurable (high-confidence error rate, **Not sure** frequency), or do participants all guess at the same rate?
4. Does usability friction (scan failures, card confusion, UI navigation) dominate the learning signal, or can participants interact with the system as intended?
5. Which distractors are effective, which feedback messages help, and which instruments are confusing? These content-refinement findings are transferable to an SPD module.

If these fail in novices, they fail everywhere, because they are properties of the measurement instrument, not of the participant population. An SPD pilot run before the measurement tool is validated produces ambiguous results: a null finding could mean the system does not help, or it could mean the scoring rubric, error categories, or convergent measure are inadequate. The novice study decouples these two failure modes.

A corollary is that the novice study does **not** establish the effect size for SPD workers. The dependent variable shifts from absolute accuracy gain (novices) to error-type reduction, confidence calibration, and

retention maintenance (workers) (Section 13.4). The novice results warrant a worker study by showing the instrument functions; they do not substitute for a worker study.

13.4 Study Ladder

Table 30: Study ladder by participant type and validity source

Study	Participants	Validity source	Kirkpatrick level
Study 0: marker reliability	1-2 project members, no learning claim	Measurement reliability (prerequisite)	–
Study 1: novice simulated learning	8-12 college students or classmates with no SPD experience	Response process, internal structure, relations to other variables (convergent)	Level 2 (Learning)
Study 2: delayed retention	Same Study 1 participants, 7 days later	Relations to other variables (temporal)	Level 2 (Learning, durability)
Study 3: expert review	1 SPD educator, instructor, or knowledgeable reviewer	Content	Level 2 (Learning, content plausibility)
Study 4: future SPD pilot	SPD trainees or technicians	Consequential, extrapolation to target context	Level 3 (Behavior)

The project should prioritize Studies 0-2 and, if possible, one expert review.

13.5 Study 0: Marker Reliability

Purpose: prevent app or camera failures from contaminating learner scores. Setup:

- Print all AprilTag cards used in the tray module.
- Use the same device, camera angle, lighting, tray boundary, and table setup planned for the user study.
- Test cards individually, in valid tray layouts, and in cluttered layouts with distractors.

Record total cards visible, cards detected, false positives, missed cards, duplicate or unstable detections, and lighting or occlusion notes. Acceptance target before user testing:

- at least 95% card detection in normal layouts;
- no systematic confusion between marker IDs;
- visible confirmation screen before final scoring so camera misses are not counted as learner errors.

13.6 Study 1: Novice Simulated Learning

Recruit 8-12 college students or classmates with no formal SPD experience and no prior exposure to the module. Collect optional background variables such as medical, biology, lab, tool, or surgical-instrument familiarity. This matches the NeuroVase-style proxy choice: students are not SPD workers, but they are acceptable for a first novice learning and usability study if the claim is bounded to simulated learning. Use a within-subject pre/post design:

intake -> pre-test -> training -> practice -> immediate post-test -> survey/interview

Target one 60 minute session per participant:

Table 31: Study 1: novice session timeline

Segment	Time	Activity
Consent and intake	3-5 min	Explain simulated study, collect background.
Device orientation	2 min	Show how to scan and submit without teaching tray answers.
Pre-test tray sort	5-8 min	No hints, no feedback, timed.
Retrieval cards	10-12 min	Prompt-before-reveal study of required items, distractors, features, and counts.
Quiz	5-8 min	Short recall/recognition quiz with confidence or Not sure .
Practice sort	8-10 min	Same style task with immediate feedback.
Immediate post-test	5-8 min	Parallel variant, no hints, no feedback, timed.
Convergent photo-ID test	5-7 min	Paper-based photo identification of each required instrument plus 2 unseen transfer-probe instruments.
Survey and interview	5-8 min	SUS or short 5-point ease/usefulness/confidence survey plus open questions.

Primary learning measures:

- tray accuracy: required items correctly selected with correct count;
- severe errors: missing, wrong, misidentified, extra, wrong-count;
- duration from task start to final submission;
- high-confidence errors;
- **Not sure** selections or low-confidence correct answers;
- weak-item recovery from quiz/practice to post-test.

Convergent evidence measures:

- **Photo-identification test:** after the post-test tray sort, learners complete a short paper-based test showing high-resolution photos of each required instrument and 2-3 distractor instruments. For each photo, they write the instrument name, indicate whether it belongs in the tray, and rate their confidence. This test shares the knowledge domain (instrument identity, aliases, lookalike discrimination) with the tray sort while differing in response format (identification without physical card selection). The target correlation between post-test tray accuracy and photo-ID accuracy is $r \geq 0.50$, following the convergent-validity standard that two measures of the same construct should show moderate-to-strong association [110, 111, 116].
- **Transfer probe:** the photo-ID test includes 2 instruments from a different tray module that the learner has never seen. If TrayGuard teaches general instrument-discrimination skill rather than module-specific memorization, learners who performed well on the trained tray should show better discrimination on unseen instruments. This is a weak test (novices have no basis to identify unseen

instruments), but the pattern across participants is informative: even chance-level performance on the transfer probe, when combined with strong within-module gains, supports the bounded claim of module-specific learning.

Usability and experience measures:

- completion rate;
- help requests;
- scan failures or rescan attempts;
- SUS or short 5-point ease/usefulness/confidence survey;
- open feedback on confusing cards, confusing instruments, and perceived realism.

Prior user studies support this design pattern. NeuroVase is the closest interaction precedent because it tested a tangible cue-card and tablet-based AR medical-learning system with 40 participants, a traditional-learning comparison, pre/post knowledge testing, SUS, Likert experience measures, and open feedback [80]. That supports TrayGuard’s use of college students as target-adjacent novice learners, provided the claim remains about simulated learning and usability rather than target-worker competence. Medical simulation app studies also commonly use novice learner cohorts when the goal is early validity, usability, or learning evidence. Tulipan et al. evaluated Touch Surgery’s carpal-tunnel-release module with medical students, orthopedic residents, and expert hand surgeons; medical students served as the novice cohort, participants completed repeated simulation attempts, performance was logged, and the novice group completed a Likert satisfaction/face-validity questionnaire [88]. The study is relevant because TrayGuard similarly expects repeated attempts, objective scoring, and user perception data to show whether the simulation is usable and whether performance changes with practice. Orthopedic VR training studies provide a second proxy-participant precedent: Orland et al. studied first- and second-year medical students as novices in a controlled orthopedic training task, comparing simulation-supported preparation with conventional preparation and measuring downstream task performance [91]. This supports using students when the target is an early, controlled, preclinical learning task, while still reserving real workplace transfer for later target-user studies. Finally, Ofstead et al. is the target-domain model rather than the proxy-user model. Their sterile-processing training pilot used certified SP professionals, pre/post testing, hands-on practice, confidence and satisfaction capture, workplace homework, and a delayed booster [28]. TrayGuard’s student study should therefore be described as a first-stage novice simulation study, with a later SPD educator review and SPD trainee pilot needed before making target-worker claims.

A stronger later design would compare flashcard-only practice against flashcards plus simulated tray sorting in a between-group design:

Table 32: Between-group study design for isolating practice effect

Group	Training condition	Post-test
A	Structured flashcards and quizzes only.	No-hints simulated tray-verification task.
B	Structured flashcards and quizzes plus simulated tray-verification practice with feedback.	Same no-hints simulated tray-verification task.

The main outcome would be whether Group B performs better on the applied tray task, measuring tray-verification accuracy, completion time, missing/extra/wrong/misidentified/wrong-count errors, high-confidence errors, low-confidence correct answers, and weak-item recovery. This design would isolate the incremental effect of simulated practice beyond retrieval cards alone. For the first project pro-

tototype, the within-subject pre/post design provides simpler logistics and smaller sample requirements; the between-group comparison is a natural follow-up once the prototype and protocol are stable.

13.7 Study 2: Delayed Retention

Same-day post-testing supports immediate learning only. It should not be called retention [105]. The delayed check distinguishes immediate practice gain from retained learning. In Kane’s validity argument framework, the extrapolation inference from test score to real-world competence depends on evidence that performance persists after a delay during which transient factors (memorization of exact layouts, test familiarity, short-term motivational effects) have decayed [115]. Without a delayed measure, the interpretive argument supports only a same-day practice-effect claim.

The primary retention interval is **7 days** from the initial session. This interval is a practical compromise: Bell et al. found that medical-knowledge gains from a single intervention could decay measurably within 7 days [105], and Larsen et al. showed that the testing-effect advantage over restudy persisted at least 7 days in a medical-education randomized trial [75]. Ofstead et al. used a longer (2-month) interval with a booster session, but their intervention included workplace homework between sessions [28]. For a single-session training tool without between-session homework, 7 days is the shortest credible retention interval that separates durable learning from ephemeral practice gain.

Procedure:

1. Ask whether the participant studied the material since the first session.
2. Run a no-hints delayed tray sort with a third parallel variant.
3. Capture confidence or **Not sure**.
4. Run the photo-identification test again to measure convergent-retention correlation.
5. Ask what they remembered and what decayed.

Falsification criteria. The retention claim is supported only if:

- delayed accuracy retains at least half of the immediate pre/post gain (i.e., $\text{delayed} - \text{pre} \geq 0.5 \times (\text{post} - \text{pre})$);
- delayed accuracy does not return to within 5 percentage points of the pre-test baseline (i.e., the retention effect is not attributable to measurement noise or test-retest practice);
- at least 70% of Study 1 participants complete Study 2.

If these criteria are not met, the project claim reverts to immediate learning only and the retention gap is reported as a limitation that future work (addressable by distributed practice, multiple training sessions, or a booster) must close. This follows Kane’s principle that the interpretive argument must include explicit rebuttals: the conditions under which the claim would not be supported [115].

Optional extended intervals. If the 7-day check succeeds and schedule permits, additional checks at 21-28 days or 2 months would strengthen the retention claim further. The 2-month interval would match the Ofstead booster precedent [28].

13.8 Study 3: Expert Review

Expert review addresses the main weakness of college-student testing: content validity. The reviewer can be an SPD educator, SPD supervisor, sterile-processing instructor, surgical technology instructor, or faculty member with relevant instrument knowledge. Materials to show:

- tray list;

- instrument cards;
- distractor list;
- aliases and distinguishing features;
- scoring rubric;
- sample learner report.

Questions:

1. Are the instrument names and aliases plausible?
2. Are the required counts and distractors plausible for a training tray?
3. Are the lookalike pairs educationally meaningful?
4. Are any feedback messages misleading or unsafe?
5. Would the weak-item report help target coaching?
6. What would need to change before using this with SPD trainees?

13.9 Analysis and Acceptance Bar

For 8-12 participants, report descriptive paired results:

- participant-level pre/post table;
- mean and median accuracy change;
- mean and median duration change;
- total error counts by type before and after;
- high-confidence errors before and after;
- weak-item examples;
- representative qualitative feedback.

Avoid strong inferential claims unless the sample is larger. If the sample is large enough, add paired differences with confidence intervals. The training claim is supported if most learners show:

- higher post-test accuracy;
- equal or lower post-test duration without accuracy decline;
- fewer severe errors, especially missing and misidentified items;
- fewer high-confidence errors;
- fewer **Not sure** responses or low-confidence correct answers;
- evidence that missed or uncertain items improve after repeated retrieval;
- interview evidence that they can name what they learned or what remains hard.

Best project claim wording:

In a novice college-student sample, TrayGuard improved immediate simulated tray-sorting performance after retrieval-centered practice. A delayed 7-day check tested short-term retention. A convergent photo-identification test provided evidence that the tray-sort scores reflect instrument knowledge rather than task-specific familiarity. Expert review was used to

assess whether the tray content and feedback were plausible for future SPD trainee validation. These findings establish that the learning mechanism works in this task domain for a naive population. They do not establish the effect size for experienced SPD workers, whose baseline on familiar trays may be high and for whom the dependent variable shifts from accuracy gain to error-type reduction, confidence calibration, and retention maintenance (Section 10-1).

13.10 Validity Claims Summary

The following table maps each project claim to the type of validity evidence that supports it, following the Cook and Beckman five-source framework [111] and Kane’s argument-based approach [115]. No single study provides all sources; the overall claim rests on the accumulated pattern.

Table 33: Validity claims mapped to evidence sources

Claim	Validity source	Evidence	Would be falsified by
The tray module content is plausible for training.	Content	Study 3 expert review; instructor-verified module; literature grounding and real count-sheet fields (Section 10.8).	Expert finds names, counts, distractors, or feedback unrealistic; module uses content that contradicts published tray listings.
Learners interact with the system as intended; confidence reflects genuine self-assessment.	Response process	Study 1 usability observation; confidence-calibration analysis (high-confidence error rate, Not sure frequency); SUS scores; interview feedback about confusion points.	Frequent scan failures produce scores that reflect app error rather than learner knowledge; confidence ratings do not distinguish correct from incorrect items; SUS scores below 68.
Error categories behave consistently and decompose learning gains.	Internal structure	Study 1-2 pre/post error-type analysis; lookalike pairs produce proportionally more misidentified errors than non-lookalike items; practice reduces specific error categories rather than shifting errors across types.	Error types do not differ between lookalike and non-lookalike pairs; accuracy gain comes entirely from one error category (e.g., fewer missing items) while other categories remain unchanged.
Tray-sort scores measure instrument knowledge and not only task-specific familiarity.	Relations to other variables	Study 1 convergent photo-ID test; correlation between tray-sort accuracy and photo-ID accuracy; transfer probe on unseen instruments.	Photo-ID correlation $r < 0.30$; photo-ID scores show ceiling or floor effects that prevent discrimination; transfer-probe performance is negatively correlated with trained-tray performance.
Learning persists beyond the initial session.	Relations to other variables (temporal)	Study 2 delayed tray sort at 7 days; retention relative to pre-test baseline; comparison of delayed accuracy to immediate post-test accuracy.	Delayed accuracy returns to within 5 pp of pre-test; less than half of the immediate gain is retained; fewer than 70% of Study 1 participants return.
The report supports instructor coaching without supporting overclaim.	Consequences	Scope boundaries (Section 14); report labels frame results as simulated practice evidence; export fields include weak items and confidence calibration.	Report language uses pass/fail or clinical-readiness labels; results are used for employment decisions; learners report feeling surveilled rather than coached.
Learning gains in novices transfer to SPD workers (extrapolation).	Generalizability / extrapolation	Ofstead precedent: experienced SP workers were well below ceiling on novel tasks; mechanism (retrieval practice + feedback) is population-independent per the training literature; dependent	Worker study fails to show improvement on any worker-relevant outcome; novice-effect-size extrapolation is cited without worker data.

13.11 Future SPD Workplace Pilot

A later SPD pilot requires local count sheets, local photos or real teaching instruments, SPD educator review, technician or trainee participants, approval/privacy review, and a comparison condition such as current study materials or preceptor-led orientation. It should measure whether the system helps target users in a workplace-relevant training context. Only that later pilot can support claims about SPD trainee usefulness, supervised transfer, or operational training adoption.

The SPD pilot must change the dependent variable from the novice study. Where novices are measured on absolute accuracy gain (pre 0 to post 90), experienced workers are measured on: error-type reduction (fewer specific errors on known trays), speed improvement without accuracy loss, confidence calibration (high-confidence error rate), new-module learning rate (time to criterion on an unfamiliar tray), and retention maintenance (preventing decay on low-frequency trays). The pilot protocol must specify which of these applies and the relevant minimum viable improvement before data collection [113, 82].

14 Risk Assessment And Scope Boundaries

14.1 Formal Risk Register

Table 34: Formal risk register with causes, consequences, and mitigations

Risk	Cause	Consequence	Mitigation	Residual Boundary
Student proxy overclaim	Participants are not SPD workers.	Reviewers reject relevance to target users.	State that students test novice simulated learning; add expert review; reserve SPD claims for future pilot.	No SPD worker effectiveness claim.
Same-day retention mistake	Immediate post-test follows practice.	Report overstates durability.	Add committed 7-day delayed check with falsification criteria; otherwise call it immediate learning only.	Retention claim only after delayed test.
Card physical realism	Cards do not reproduce weight, scale, tactile handling, or real visual lookalikes.	Transfer to real instruments remains unknown.	Use cards as selection/count proxy; add local photos or real teaching instruments later.	No clinical handling competency claim.
Marker detection errors	Lighting, occlusion, angle, or overlap causes missed detections.	App errors look like learner errors.	Run Study 0; log raw detections; show confirmation before scoring.	Marker reliability must be reported.
Answer leakage from cards	Labels or marker IDs reveal the answer.	Measured gain is not learning.	Use neutral assessment fronts; randomize marker IDs; hide answer text in assessment.	Study and assessment cards must be separated.
Content invalidity	Student-built module may use wrong names, counts, or distractors.	Results do not reflect plausible SPD training.	Require expert review of module content and feedback.	Content validity limited until SPD review.
Feedback contaminates assessment	Practice answers leak into post-test.	Pre/post comparison inflated.	Use parallel variants; suppress hints and feedback in assessment modes.	Cannot claim broad generalization from one tray.
Usability friction dominates learning	Scanning or workflow confusion slows participants.	Poor scores reflect interface friction.	Track help requests, scan failures, SUS/ease ratings, and qualitative feedback.	Usability results must be reported separately.

14.2 Unknowns and Concerns

The current design is viable as a low-cost formative learning prototype, but several concerns remain unresolved:

Table 35: Unknowns and concerns with strengthening actions

Concern	Why it matters	Strengthening action
Card-to-instrument transfer	AprilTag cards preserve selection and counting decisions, but they do not reproduce weight, scale, handling, tactile inspection, or real visual lookalikes.	Treat the first study as simulated learning only; add an SPD educator review and a later task using local photos or real teaching instruments.
Flashcard overreach	Flashcards support recall, but tray work requires applying local rules under distractor pressure.	Keep flashcards as preparation, then make the main outcome a no-hints tray sort with missing, extra, wrong, misidentified, and wrong-count scoring.
Marker identity leakage	If cards visibly encode names or IDs, learners may match labels instead of learning instruments.	Use neutral card fronts during assessment, randomize marker IDs, hide answer text, and separate study cards from assessment cards.
Content validity	A student-built tray module may contain wrong names, unrealistic distractors, or nonlocal quantities.	Require instructor or SPD educator verification of every instrument, alias, count, and distractor before using results as evidence.
Assessment equivalence	Pre/post gains are weak evidence if the post-test is easier or repeats the same exact layout.	Build parallel pre/post tray variants with matched item families, counts, distractors, and difficulty.
Short-term learning only	A same-day post-test can show practice effects rather than retention.	Add a delayed retention check or booster if schedule permits.
Novice participant limits	Student participants can test learnability, but they do not represent SPD technicians under workflow pressure.	Report student results as novice simulated-learning evidence and reserve adoption/clinical-transfer claims for an SPD pilot.
Scanning reliability	Camera angle, lighting, occlusion, and card overlap may produce app errors that look like learner errors.	Log raw detections, show a confirmation state before scoring, and run a marker-detection bench test before the user study.
Confidence calibration	Confidence ratings can become noise if captured too often or too vaguely.	Capture simple confidence or Not sure at assessment moments and analyze high-confidence errors separately.
Instructor usefulness	Better learner scores do not automatically mean the output helps educators.	Add a short instructor-facing report and ask an educator whether the weak-item summary would change coaching.

These holes do not invalidate the design. They define the claim boundary: the prototype can show whether a marker-card training loop improves novice performance on a simulated local tray task. It cannot yet show real SPD competence, reduced OR delays, or deployment-ready instrument recognition.

14.3 Scope Boundaries

Allowed after Study 1:

- novices improved on a simulated tray task;
- the interface was usable for first-time learners;
- retrieval cards plus practice sorting produced measurable immediate gains.

Allowed after Study 2:

- gains persisted after 7 days (meeting the falsification criteria in Section 13.7);
- the system showed measurable short-term retention in a simulated tray task, distinguishable from same-day practice effects.

Allowed after Study 3:

- an expert reviewer found the module plausible for future SPD validation;
- content concerns were identified and revised before broader testing.

Not allowed yet:

- SPD technicians will improve;
- the system reduces tray errors in hospitals;
- the system reduces OR delays;
- the system certifies competency;
- the app recognizes real surgical instruments;
- same-day gains demonstrate retention.

15 Broader Impacts And Ethics

TrayGuard’s positive impact is a more accessible way to practice local tray knowledge before a learner consumes scarce supervised training time. If it works, learners get repeatable practice, instructors get clearer weak-item evidence, and future teams get a safer bridge between training content and camera-supported tray review. The ethical risk is that a useful training report could be misused. A learner score is not a certification result, employment screen, disciplinary record, or permission to perform unsupervised sterile-processing work. The report should therefore label outputs as simulated practice evidence and should avoid language such as pass, fail, competent, or cleared unless a formal program adds its own supervised competency process. Using the principles of beneficence, nonmaleficence, autonomy, and justice:

Table 36: Bioethics principles applied to TrayGuard

Principle	TrayGuard implication
Beneficence	The system should help learners improve and help educators target coaching, especially on missing, wrong, misidentified, and high-confidence errors.
Nonmaleficence	The system should not imply clinical readiness, replace hands-on supervision, or hide uncertainty behind a single score.
Autonomy	Learners and instructors should understand what is being recorded, why it is being recorded, and how the report will be used.
Justice	Access should not depend on having a perfect local instrument library, expensive hardware, or one preferred training site. Modules should be updateable and printable.

Negative externalities and controls:

Table 37: Negative externalities and ethical controls

Risk	Ethical concern	Control
Punitive learner metrics	Scores could become surveillance or discipline instead of coaching.	Use hashed IDs for studies; report weak items and practice progress; prohibit employment or competency decisions from prototype data.
Overreliance on simulated scores	A high score could be mistaken for real SPD readiness.	Label all results as simulated; require supervised hands-on validation for clinical claims.
Reduced supervised practice	Programs might use the tool to cut preceptor time.	Frame TrayGuard as preparation for supervised practice, not replacement.
Unequal local content	Better-resourced sites may build richer modules.	Use open file formats, printable cards, and a minimal module template.
Instrument-library bias	Learners may only practice instruments represented in the module.	Version modules, list coverage gaps, and require local review before use.
Answer leakage	Labels or marker IDs could inflate learning results.	Use neutral assessment cards and hide names/counts in assessment modes.
Printing and material waste	Card iterations create paper/plastic waste.	Use reusable sleeves, print small modules, and revise digitally before reprinting.
Privacy creep	Future reports could collect names, staff IDs, or workplace behavior.	Keep patient data out of scope; collect only fields needed for learning analysis.

These controls are part of the design, not optional documentation. They protect the stakeholder value of TrayGuard by keeping the training tool honest about what it can and cannot prove.

16 Project Plan And Future Work

16.1 Completed Work

The following subsystems have been implemented:

- **Training web app.** A FastAPI backend serves a single-page browser application with seven-step learner flow: intake, pre-test, study cards, quiz, practice sort, post-test, and summary. Camera capture through `getUserMedia` streams frames to a server-side ArUco detection endpoint. Manual quantity entry serves as a fallback when camera access is unavailable or unreliable `src/trayguard/web/app.py`.
- **Tray module data model.** Eight file-backed tray modules define required instruments, distractors, lookalike pairs, assessment variants, and ArUco marker mappings. The primary module, `basic_general_tray_v1`, contains 10 required instrument types (15 total required units), 5 distractors, 6 lookalike pairs, and 2 assessment variants with matched difficulty. Five instrument catalogs covering 400+ classes provide authoring reference material `config/tray_modules/`.
- **Card generation and printing.** A CLI command generates printable ArUco marker PNGs, a marker-grid index sheet for camera scanning, and double-sided study flashcards with instrument images, names, aliases, distinguishing features, and tray-specific quantities. An image-processing pipeline handles EXIF rotation, content-aware cropping, and automatic marker placement `src/trayguard/learning/cards.py`.
- **Scoring engine.** A scoring function classifies each item into correct, missing, extra, wrong, misidentified (lookalike substitution), or wrong-count categories. Error-point aggregation maps to an accuracy score and required-recall metric. Confidence ratings (1-5 scale and a `Not sure` checkbox) are captured per item and flagged as high-confidence errors or low-confidence correct answers `src/trayguard/learning/scoring.py`.
- **Export and run storage.** Each completed run produces a JSON payload plus CSV files at the attempt level and item level. Learner identifiers are hashed (SHA-256 truncated to 16 hex characters). The export schema includes all error categories, confidence metadata, and module version fields `src/trayguard/learning/export.py`; `src/trayguard/learning/run_store.py`.
- **CV data collection and annotation pipeline.** A webcam capture tool with bounding-box annotation UI supports rapid multi-session data collection with per-class image output, YOLO label export, and session save/load `src/trayguard/data_collection/`.
- **YOLO training infrastructure.** A Hydra-configured training pipeline supports model, dataset, and hyperparameter overrides across YOLO11n/s/m/l variants and RT-DETR-l. Training runs are deterministic and reproducible through a fixed seed `src/trayguard/training/`.
- **CV baseline: Lavado surgical instruments.** A YOLO11s model was trained on the Lavado dataset (2,106 images, 4 classes) and validated on a held-out test set, achieving mAP50 of 0.990 and mAP50-95 of 0.944 `data/lavado/`.
- **CV experiments: lighting and occlusion robustness.** Seven staged brightness experiments (800–80 lumens) on matte, reflective, separated, and overlay layouts were trained and validated. Per-stage precision, recall, mAP50, mAP50-95, and per-image count-error metrics were collected across 11 brightness levels and both backgrounds `scripts/validate_brightness_experiments.py`.
- **CV experiments: lighting diversity.** Three staged spoon lighting experiments tested whether adding lighting variety during training improves held-out-condition detection across reference, 45-degree, and 90-degree flashlight conditions `data/spoon_lighting_yolo/`.

- **Multi-variant benchmark harness.** A benchmark harness supports YOLO11n/s/m/l at 640 and 960 image sizes and RT-DETR-l at 640 for precision, recall, mAP, and latency comparison `src/trayguard/training/benchmark.py`.
- **Legacy Streamlit tray-check demo.** A live YOLO tray-check interface demonstrates camera capture, detection, and tray-status comparison for proof-of-concept review `src/trayguard/app/streamlit_app.py`.
- **Tests.** Unit tests verify module loading, scoring accuracy (full-credit and lookalike-substitution cases), CSV/JSON export file completeness, and ArUco marker generation and detection `tests/test_learning.py`.

16.2 Project Plan Postmortem

The original project direction emphasized computer vision support for tray checking. That direction remained technically plausible, but the validation burden was too large for the project schedule: real instrument recognition would require local instrument data, site-specific tray rules, cross-manufacturer generalization testing, camera reliability, clinical workflow review, and adoption evidence. The project therefore pivoted to a training-centered system with a stronger near-term evidence path. The pivot reduced risk by changing the main proof obligation from deployment-grade recognition to measurable novice learning. The remaining engineering work is still concrete: build a local tray module, use AprilTag cards to make the simulated physical task observable, score tray attempts, and run a small pre/post study with claim boundaries.

16.3 Future Work Roadmap

Table 38: Future work roadmap phases

Phase	Goal	Exit Criteria
1. Minimal prototype	Build the end-to-end local tray loop.	Load module, scan cards, run pre-test, show study cards, run quiz, run practice sort, run post-test, export metrics.
2. Verified module	Make the first tray content defensible.	8-12 required instruments, distractors, lookalikes, neutral assessment cards, matched variants, and expert/instructor review.
3. Marker reliability	Prove the app can observe card choices.	At least 95% detection in normal layouts, no systematic ID confusion, confirmation screen before scoring.
4. Novice study	Test immediate simulated learning.	8-12 college-student participants, pre/post results, usability notes, and bounded claim language.
5. Retention check	Test whether learning persists.	7-day delayed no-hints tray sort with third matched variant, plus falsification criteria (Section 13.7).
6. Report refinement	Turn study output into project evidence.	Participant-level results, aggregate pre/post plots, error-category table, high-confidence errors, and qualitative themes.
7. SPD educator review	Improve content validity.	Reviewer feedback on names, counts, distractors, lookalikes, scoring, and weak-item report.
8. Future SPD pilot	Test target-user relevance.	SPD trainee or technician participants, local count sheet/photos, approval/privacy review, and comparison condition.
9. Optional CV support	Reintroduce real-instrument recognition as support, not the core proof.	Detector tested on local photos with generalization and failure-mode reporting.

16.4 GitLab Epic / Issue Map

Table 39: Epic and issue map for continuation

Epic	Issues
Training Module	Define first tray module; add instrument cards with aliases and features; add tray quantities and distractors; build study-card view; build identification quiz.
Scoring And Feedback	Record quiz correctness, duration, and confidence; build simulated tray sorting; implement missing/extra/wrong/misidentified/wrong-count scoring; add practice feedback.
Assessment	Build pre-test mode; build post-test mode; suppress hints and feedback during assessment; add delayed-retention mode if schedule allows.
Evidence And Reporting	Export learner metrics; generate pre/post summary; report confidence calibration, high-confidence errors, and weak-item recovery.
Pilot Evidence	Write pilot walkthrough script; run marker reliability check; run novice study; record confusion points; add limitation note about non-SPD participants.
Content Validation	Prepare expert-review packet; collect reviewer feedback; revise names, counts, distractors, and feedback language.
Presentation	Lock final demo narrative around simulated learning, retention boundary, and future SPD validation.

16.5 Next Team Priorities

16.6 Budget

Table 40: Project budget by category

Category	Expected Cost	Notes
Software prototype	\$0 direct cost	Uses existing local app stack and open-source AprilTag support where available.
Physical cards	\$5-25	Printer paper/cardstock, sleeves, tape, or foam backing.
Tray surface	\$0-30	Existing table, poster board, foam board, or simple marked tray boundary.
Camera/device	\$0 if using existing laptop/tablet/phone	Optional stand improves reliability but is not required for the first study.
Participant cost	\$0-100	Small snacks/gift cards if allowed; class demo can be unpaid.
Expert review	\$0-200	Likely unpaid instructor feedback; paid SPD educator review would be stronger.
Future SPD pilot	TBD	Depends on site approval, staff time, privacy review, and local instrument access.

The main cost is labor: module authoring, prototype implementation, marker testing, study moderation, analysis, and report writing.

17 Discussion And Final Recommendations

This report supports a clear next step: build and test the software-first training loop before investing more effort in deployment-grade tray automation. The problem evidence is strong enough to justify the target task, and the learning evidence is strong enough to justify a simulated training prototype. The current evidence is not strong enough to claim that TrayGuard reduces hospital tray errors, improves SPD technician performance, or recognizes real surgical instruments in clinical conditions.

The evaluation plan now follows established frameworks from the training- evaluation and educational-assessment literatures. Kirkpatrick’s four-level model frames the project’s reach: the current studies target Level 2 (Learning), with internal-structure evidence from error-type decomposition, convergent evidence from an independent photo-identification test, and temporal evidence from a committed 7-day retention check. Level 3 (Behavior transfer) and Level 4 (Results) remain future work requiring SPD-site participants and operational outcome data [113]. The five-source validity framework from Cook and Beckman, grounded in Messick’s unified validity theory and Kane’s argument-based approach, provides the evidentiary structure for each claim level, with explicit falsification criteria for the retention and convergent-validity claims [110, 115, 111, 114].

A further validity concern is population generalizability. The novice study tests complete novices (college students). Experienced SPD workers may already perform at or near ceiling on their current trays. A pre/post gain from 0 to 90 in novices does not predict a gain from 90 to 90 in workers. The evaluation design addresses this by separating the dependent variable by population: novices are measured on

absolute accuracy gain; workers would be measured on error-type reduction, confidence calibration, new-module learning rate, and retention maintenance (Section 13.4). The Ofstead precedent shows that even certified SP workers have substantial room for improvement on novel tasks (41% pre-test in their study), supporting the claim that TrayGuard’s value for an experienced population lies in targeted error reduction rather than general accuracy gain. The novice studies establish that the learning mechanism works in this task domain; the SPD pilot is required to establish the effect size for the target population.

Stakeholders can conclude the following:

- Tray reconstruction is a defensible focus because many readiness failures involve assembly, visualization, identification, missing items, wrong items, extra items, and local count-sheet use.
- The training pivot is justified because local tray knowledge and supervised practice are real bottlenecks, while deployment-grade CV has a much larger validation burden.
- A marker-card prototype can support a bounded novice-learning study if the cards are neutral, detection is reliable, assessment variants are comparable, and results are labeled as simulated.
- The project has reduced concept risk, requirement risk, scoring risk, and study design risk by defining the module contract, performance targets, error categories, study ladder, convergent validity target, and claim boundaries.
- The evaluation design addresses content validity (expert review), internal structure (error-type analysis), relations to other variables (convergent photo-ID test), and temporal persistence (committed 7-day retention check with falsification criteria).

Stakeholders cannot yet conclude that TrayGuard:

- certifies sterile-processing competence;
- replaces hands-on hours, preceptor observation, or local sign-off;
- reduces OR delays or hospital tray-defect rates;
- works for SPD technicians in a real workplace;
- recognizes real instruments across manufacturers, lighting, occlusion, and tray layouts;
- produces the same effect size for experienced workers as for novices (the dependent variable and expected magnitude differ by population);
- supports retention unless the 7-day delayed check meets the falsification criteria specified in Section 13.7.

The next team should therefore implement the minimum evidence loop:

```
load module -> pre-test -> study cards -> quiz -> practice sort -> post-test -> convergent photo-ID test
-> export report
```

After that loop works, the team should run the marker reliability check, then a small novice pre/post study with the convergent photo-ID measure, then the committed 7-day retention check, then an expert review of names, counts, distractors, lookalikes, and feedback. Only after those results should the project expand toward real teaching instruments, local photos, SPD trainees, or real-instrument CV. Final recommendation:

Treat TrayGuard as a local tray learning and assessment platform first. Use the project prototype to prove that novices can improve on a simulated local tray task, demonstrate that the gains reflect instrument knowledge (convergent photo-ID evidence), persist for at least 7 days (retention check), and are grounded in plausible content (expert review), then use

those results to decide whether a future SPD pilot or camera-supported tray review system is worth the additional validation cost.

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A Research Support Audit

Table 41: Research support audit: design choice traceability

Design Choice	Research Support	Defensible Stance
Time-to-competency as the north star	SPD certification and training evidence shows that entry into practice requires hands-on hours, structured training, and preceptor support [27]; [117]; [29].	Measure simulated improvement in local tray familiarity and report the longitudinal onboarding question as future field validation.
Retrieval-centered learning loop	Simulation, retrieval practice, spacing, feedback, and pre/post testing are supported by health-professions and learning-science evidence, with direct SP training precedent in Ofstead et al. and digital-practice support from spaced education and electronic-flashcard research [28]; [71]; [74]; [75]; [77]; [78]; [79].	Build pre-test, retrieval-first cards, quizzes, applied tray practice with feedback, weak-item review, post-test, and metrics export as one module.
Local tray authoring	Instrument families are reusable, but actual tray requirements are local: count sheets specify tray contents, quantities, sizes, and catalog/reference numbers; tray optimization depends on procedure, surgeon preference, usage likelihood, and stock decisions; specialty and loaner trays can be vendor-specific [11]; [14]; [23].	Start with file-backed instructor-verified modules and record local names, aliases, quantities, photos, and version history.
Human authority	Healthcare AI literature warns about automation bias, deskilling, liability, and workflow fit [94]; [118]; [95]; [119].	The system may teach, score, explain, and log; instructors and supervisors remain responsible for real competency judgments.
Workflow-first learner UI	General usability guidance supports visible status, efficient action paths, error prevention, and recoverable mistakes. Human-AI guidance supports efficient invocation, dismissal, and correction when AI guesses wrong. Clinical alert literature warns that low-value alerts and added tasks can undermine acceptance [120]; [121]; [122].	Design around the learning mode, use clear mode boundaries, provide fast practice feedback, and suppress hints during assessment.
CV feasibility as an external assumption	Few-shot object detection and surgical-instrument CV papers show that data-efficient detection and constrained instrument recognition are plausible [57]; [55]; [56].	Cite existing papers for feasibility and reserve local CV accuracy for future support-layer validation.
CV generalization as a future validation problem	Kienle et al. show a large cross-manufacturer performance drop despite strong in-domain instrument-stand results [58].	Any future camera-supported feature needs site-specific validation. The current project can function through manual simulated sorting.
Lookalike tools as learning content	HOSPITools and Atabuzzaman et al. highlight subtle instrument differences. Also, distinguish if	Teach distinguishing features in retrieval-first cards and track individualized performance

B Implementation Defaults

Table 42: Implementation defaults and critique-ready interpretation

Default	Current Use	Critique-Ready Interpretation
File-backed tray and instrument data	Fast local authoring for one seeded module.	Good for a capstone project. A real training program would need role permissions, content review, versioning, and integration decisions.
Pre/post simulated tray sorting	Primary training-effectiveness evidence.	Measures simulated local familiarity and produces paired learning metrics.
Confidence ratings	Learner metacognition and instructor review.	Useful for spotting overconfidence and fragile knowledge.
Existing YOLO/Streamlit tray-check code	Demo code and possible future visual support.	Useful as support-layer infrastructure after separate validation.
Local ignored <code>data/</code> , <code>runs/</code> , and <code>weights/</code> folders	Keeps raw images, training outputs, and weights out of git.	Sensible prototype hygiene. Formal deployment would require access controls, retention policy, audit policy, and privacy review.